Fluid Mechanics Notes

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April 2, 2026

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1 Introduction and Flow Kinematics

1.1 Streamlines, pathlines and streaklines

A **streamline** is a curve that is locally tangent to the velocity field at every point. This requirement can be expressed as:

$$dx/u = dy/v = dz/w \tag{1.1.0.1}$$

Streamlines can be found by using this relation to establish one or more differential equations.

A **pathline** is the curve obtained by following the trajectory of a single particle over time. To find the pathlines, use this relationship between the derivatives of the position of the particle and the velocity field

$$\frac{d}{dt}\mathbf{x}_{\text{particle}} = \mathbf{u} \tag{1.1.0.2}$$

And integrate with the initial conditions $\mathbf{x}(t_0) = (x_0, y_0, z_0)$ to get

$$\mathbf{x}_{\text{particle}}(t) = \mathbf{x}_{\text{particle}}(t, x_0, y_0, z_0, t_0) \tag{1.1.0.3}$$

A **streakline** is the curve obtained by connecting all fluid particles that will ever pass through a given point in space. The equations for streaklines are obtained in a similar way to those for pathlines

$$\mathbf{x}_{\text{streakline}}(t) = \mathbf{x}_{\text{streakline}}(t, x_0, y_0, z_0, t_0) \tag{1.1.0.4}$$

The important distinction between eqs. (1.1.0.3) and (1.1.0.4) is that t_0 is kept fixed in (1.1.0.3), since we are only considering one particle (the one that is released at t_0). In (1.1.0.4), we vary t_0 to obtain the trajectories of all particles that will ever pass through (x_0, y_0, z_0).

1.2 Lagrangian and Eulerian descriptions

- **Lagrangian:** follows the trajectory of a *specific particle* over time (particle description of motion)
- **Eulerian:** describes the motion of an arbitrary particle *at a particular location and specific instant of time* (field description of motion)

Motivation for the Eulerian description:

The Eulerian approach solves the question: *Given a velocity field* $\mathbf{V}(x, y, z, t)$*; find the acceleration* $\mathbf{a}_p$ *of a fluid particle at* (x, y, z, t).

We start by noting that the instantaneous velocity of the particle is given by the velocity field at (x, y, z, t)

$$\mathbf{u}_p(t) = \mathbf{V}(x, y, z, t) \tag{1.2.0.1}$$

At $t + dt$ the particle will have a new instantaneous velocity

$$\mathbf{u}_p(t + dt) = \mathbf{V}(x + dx, y + dy, z + dz, t + dt) \tag{1.2.0.2}$$

The infinitesimal change in the particle's velocity is given by the chain rule

$$d\mathbf{u}_p = \frac{\partial \mathbf{V}}{\partial x}dx + \frac{\partial \mathbf{V}}{\partial y}dy + \frac{\partial \mathbf{V}}{\partial z}dz + \frac{\partial \mathbf{V}}{\partial t}dt \tag{1.2.0.3}$$

So the total acceleration of the particle is

$$\mathbf{a}_p = \frac{d\mathbf{V}}{dt} \tag{1.2.0.4}$$

$$= \frac{\partial \mathbf{V}}{\partial x}u + \frac{\partial \mathbf{V}}{\partial y}v + \frac{\partial \mathbf{V}}{\partial z}w + \frac{\partial \mathbf{V}}{\partial t} \tag{1.2.0.5}$$

At this point is is useful to introduce the so-called *material derivative*, or *substantial derivative*

$$\frac{D}{Dt} = u\frac{\partial}{\partial x} + v\frac{\partial}{\partial y} + w\frac{\partial}{\partial z} + \frac{\partial}{\partial t} \tag{1.2.0.6}$$

$$= \frac{\partial}{\partial t} + (\mathbf{V} \cdot \nabla) \tag{1.2.0.7}$$

Which allows us to write the acceleration of the particle as

$$\mathbf{a}_p = \frac{D\mathbf{V}}{Dt} = (\mathbf{V} \cdot \nabla)\mathbf{V} + \frac{\partial \mathbf{V}}{\partial t} \tag{1.2.0.8}$$

The first term represents movement due to convection (the steady part of the flow) and the time derivative term represents the local acceleration due to unsteadiness in the flow. Eq. (1.2.0.8) is the equation used to compute the acceleration of fluid particles in the Eulerian description.

1.3 Kinematics

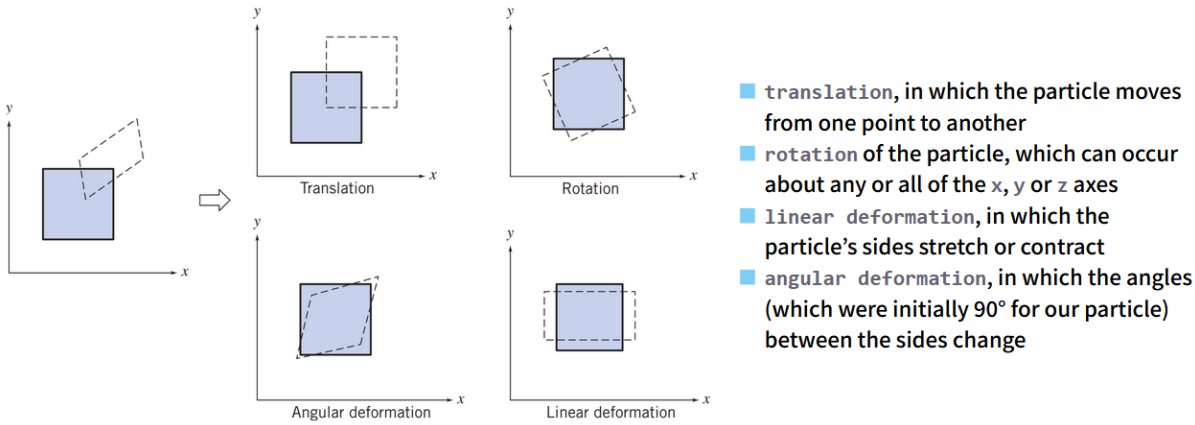


Figure 1: Various types of motion of a 2D fluid segment.

Fig. 1 illustrates the four types of motion that are possible for a 2D fluid segment. Any time evolution of a fluid segment will be a combination of these four types of motion.

1.3.1 Translation

The translation of an individual fluid particle is connected with the velocity field via eq. (1.2.0.8)

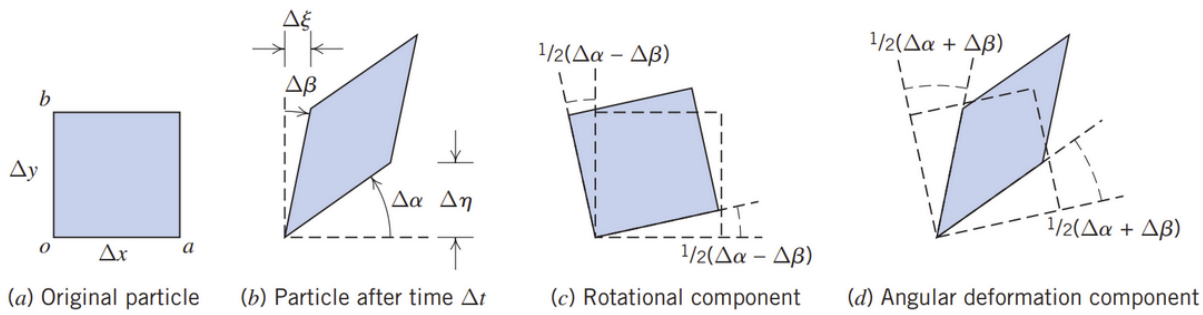
$$\mathbf{a}_p = \frac{D\mathbf{V}}{Dt} = (\mathbf{V} \cdot \nabla)\mathbf{V} + \frac{\partial \mathbf{V}}{\partial t}$$

Since this can be solved to find the translation of an individual particle, it may be used to describe how fluid segments are translated as well. We see that translatory motion depends only on the acceleration of particles in the flow.

1.3.2 Rotation

The rotation of a particle in a flow can be modelled with a vector $\omega = (\omega_x, \omega_y, \omega_<)$ whose components are the angular velocities associated with rotation about the three coordinate axes. Fig. 2 a) and b) show how a square fluid segment undergoes some deformation and rotation during a time interval Δt. The deformation is described by the translation displacements $\Delta\eta$ and $\Delta\xi$ as well as the angular displacements $\Delta\alpha$ and $\Delta\beta$. Note that $\Delta\alpha$ is positive, while $\Delta\beta$ is negative, since we use the convention that counterclockwise rotation is the positive direction of rotation.

Fig. 2 shows



(a) Original particle (b) Particle after time Δt (c) Rotational component (d) Angular deformation component

Figure 2

The segment's rigid body rotation (the rotational motion) can be found by taking the average of the two angular displacements, see fig. 2 c). The minus sign in $\frac{\Delta\alpha - \Delta\beta}{2}$ is due to the fact that we are taking the average in the counterclockwise direction, and the counterclockwise rotation of ob is $-\Delta\beta$.

We will also need to find the translational displacements $\Delta\eta$ and $\Delta\xi$. To do so, we will consider the situation where all of the displacements and angles involved in the problem are small. Then we can Taylor expand the velocity field $\mathbf{V}$ to get

$$V_x(b) = V_x(o) + \frac{\partial V_x}{\partial y}\Delta y \tag{1.3.2.1}$$

So the x-displacement of b will be $\left(V_x(o) + \frac{\partial V_x}{\partial y}\Delta y\right)\Delta t$. To find $\Delta\xi$ we must subtract the displacement of the origin which is $V_x(o)\Delta t$. That leaves us with

$$\Delta\xi = \frac{\partial V_x}{\partial y}\Delta y \Delta t \tag{1.3.2.2}$$

The same argument can be used to obtain

$$\Delta\eta = \frac{\partial V_y}{\partial x}\Delta x \Delta t \tag{1.3.2.3}$$

In our small-displacement approximation, we can write the angular displacements as

$$\Delta\alpha = \frac{\Delta\eta}{\Delta x} \tag{1.3.2.4}$$

$$\Delta\beta = \frac{\Delta\xi}{\Delta y} \tag{1.3.2.5}$$

Leading to the following result:

$$\frac{1}{2}\frac{(\Delta\alpha - \Delta\beta)}{\Delta t} = \frac{1}{2}\frac{\left(\frac{\Delta\eta}{\Delta x} - \frac{\Delta\xi}{\Delta y}\right)}{\Delta t} = \frac{1}{2}\frac{\left(\frac{\partial V_y}{\partial x}\Delta t - \frac{\partial V_x}{\partial y}\Delta t\right)}{\Delta t} \tag{1.3.2.6}$$

To find the angular velocity about the z-axis, we take the limit $\Delta t \to 0$

$$\omega_z = \frac{1}{2}\left(\frac{\partial V_y}{\partial x} - \frac{\partial V_x}{\partial y}\right) \tag{1.3.2.7}$$

This is the z-component of the curl of the velocity field (multiplied by a factor of 1/2). The same argument can be applied to fluid segments in the yz- and xz-planes, so we conclude that

$$\omega = \frac{1}{2}\nabla \times \mathbf{V} = \zeta \tag{1.3.2.8}$$

Where we've introduced the vorticity $\zeta = 2\omega = \nabla \times \mathbf{V}$ to get rid of the factor of 1/2.

1.3.3 Angular Deformation

Fig. 1 d) illustrates how the angular deformation component is the sum of the individual angular displacements $\Delta\alpha + \Delta\beta$ (find out where the factor of 1/2 comes from!)

We can repeat the last steps of our previous calculation to see that the total angular deformation in the xy-plane R_{xy} is given by

$$R_{xy} = \lim_{\Delta t \to 0} \frac{1}{2}\frac{(\Delta\alpha + \Delta\beta)}{\Delta t} \tag{1.3.3.1}$$

$$= \frac{1}{2}\left(\frac{\partial V_y}{\partial x} + \frac{\partial V_x}{\partial y}\right) \tag{1.3.3.2}$$

And by the same reasoning

$$R_{yz} = \frac{1}{2}\left(\frac{\partial V_z}{\partial y} + \frac{\partial V_y}{\partial z}\right) \tag{1.3.3.3}$$

$$R_{xz} = \frac{1}{2}\left(\frac{\partial V_z}{\partial x} + \frac{\partial V_x}{\partial z}\right) \tag{1.3.3.4}$$

So angular deformation and rotation is described by the rotation tensor $R_{ij} = \frac{1}{2}\left(\frac{\partial V_i}{x_j} - \frac{\partial V_j}{x_i}\right)$

1.3.4 Linear Deformation

Linear deformation (deformation that preserves right angles) in the x_i-direction is only possible if $\frac{\partial V_i}{x_i} \neq 0$. This quantity represents the longitudinal strain rate in the x_i-direction. Unlike other types of deformation, linear deformation may lead to changes in volume given by the volume dilation rate

$$\text{volume dilation rate} = \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z} = \nabla \cdot \mathbf{V} \tag{1.3.4.1}$$

A flow is *incompressible* if the volume dilation is zero $\nabla \cdot \mathbf{V} = 0$.

We see that linear deformation is described by the strain rate tensor $S_{ij} = \frac{1}{2}\left(\frac{\partial V_i}{x_j} + \frac{\partial V_j}{x_i}\right)$

1.3.5 Tensor Representation of Deformation

To model the deformation that fluid segments undergo, we need to consider the gradient of the velocity field (we have just seen that the derivatives of the velocity field are needed to describe the various types of fluid segment motion).

The gradient of the velocity field is a tensor with the following entries

$$\nabla \mathbf{V} = A_{ij} = \partial_i v_j \tag{1.3.5.1}$$

It is a known result from tensor calculus that *any* tensor can be decomposed into a symmetric and an antisymmetric part. This is easily shown by writing the tensor T_{ij} in a clever way and adding and subtracting the transpose T_{ji}

$$\begin{aligned} T_{ij} &= \frac{1}{2}T_{ij} + \frac{1}{2}T_{ij} \\ &= \frac{1}{2}T_{ij} + \frac{1}{2}T_{ji} + \frac{1}{2}T_{ij} - \frac{1}{2}T_{ji} \\ &= \frac{1}{2}(T_{ij} + T_{ji}) + \frac{1}{2}(T_{ij} - T_{ji}) \end{aligned}$$

It can be shown that the first term is symmetric and the second is antisymmetric. Applying this result to A_{ij}, we see that

$$A_{ij} = \frac{1}{2}(\partial_i v_j + \partial_j v_i) + \frac{1}{2}(\partial_i v_j - \partial_j v_i) \tag{1.3.5.2}$$

Considering the components of the symmetric and antisymmetric parts leads to the revelation that the symmetric part is the angular rotation tensor and the antisymmetric part is the strain rate tensor (both encountered in the previous sections)

$$A_{ij} = \frac{1}{2}E_{ij} + \frac{1}{2}S_{ij} \tag{1.3.5.3}$$

2 From Newton's Second Law to the Navier-Stokes Equations

2.1 Conservation of Mass in an arbitrary Control Volume

Important distinction: For the purposes of this section, a *material volume* is the volume occupied by a specific collection of fluid particles, while a *control volume* is any

arbitrary volume whose contents may change over time. In other words, the *material volume* changes as the fluid particles move, while the control volume can be chosen arbitrarily.

Consider a (possible time-dependent) material volume $V(t)$ in a flowing fluid with the fluid density $\rho(\mathbf{x}, t)$. Then the statement of conservation of mass can be expressed as

$$\frac{d}{dt}\int_{V(t)} \rho(\mathbf{x}, t) dV = 0 \tag{2.1.0.1}$$

Now we apply Reynolds transport theorem which makes the following statement (add derivation of this to the mathematics section)

$$\frac{d}{dt}\int_{V(t)} \rho(\mathbf{x}, t) dV = \int_{V(t)} \frac{\partial \rho(\mathbf{x}, t)}{\partial t} dV + \int_{A(t)} \rho(\mathbf{x}, t)\mathbf{b} \cdot \mathbf{n} dA \tag{2.1.0.2}$$

Note that $A(t)$ is the surface of the material volume $V(t)$. Furthermore, $\mathbf{b}$ is the velocity of A and $\mathbf{n}$ is the outward normal vector of A. Applying Reynolds transport theorem to the conservation of mass leads to

$$\int_{V(t)} \frac{\partial \rho(\mathbf{x}, t)}{\partial t} dV + \int_{A(t)} \rho(\mathbf{x}, t)\mathbf{u}(\mathbf{x}, t) \cdot \mathbf{n} dA = 0 \tag{2.1.0.3}$$

Where we have labeled the velocity of the material surface with $\mathbf{u}(\mathbf{x}, t)$, to distinguish it from the control surface we will introduce in a moment.

So far, we have been considering a *material volume*. However, a statement of conservation of mass for an arbitrary *control volume* is highly desirable. As a first step, we apply Reynolds transport theorem to the conservation of mass in the control volume V^*

$$\frac{d}{dt}\int_{V^*(t)} \rho(\mathbf{x}, t) dV - \int_{V^*(t)} \frac{\partial \rho(\mathbf{x}, t)}{\partial t} dV - \int_{A^*(t)} \rho(\mathbf{x}, t)\mathbf{b} \cdot \mathbf{n} dA = 0 \tag{2.1.0.4}$$

Since the control volume is arbitrary, we can choose V^* such that $V^*(t) = V(t)$ and $A^*(t) = A(t)$ at the specific time t we are considering. Since V^* and V are only instantaneously coincident the time derivative terms $\frac{d}{dt}\int \rho dV$ are not necessarily equal at t. However, the terms that don't contain time derivatives *are* equal at the time t. Combining this knowledge with eq. (2.1.0.4) gives us

$$\int_{V(t)} \frac{\partial \rho(\mathbf{x}, t)}{\partial t} dV = -\int_{A(t)} \rho(\mathbf{x}, t)\mathbf{u}(\mathbf{x}, t) \cdot \mathbf{n} dA = \int_{V^*(t)} \frac{\partial \rho(\mathbf{x}, t)}{\partial t} dV = -\int_{A^*(t)} \rho(\mathbf{x}, t)\mathbf{u}(\mathbf{x}, t) \cdot \mathbf{n} dA \tag{2.1.0.5}$$

We can use this to replace the second term in eq. (2.1.0.4)

$$\frac{d}{dt}\int_{V^*(t)} \rho(\mathbf{x}, t) dV - \int_{V^*(t)} \frac{\partial \rho(\mathbf{x}, t)}{\partial t} dV - \int_{A^*(t)} \rho(\mathbf{x}, t)\mathbf{b} \cdot \mathbf{n} dA = 0 \tag{2.1.0.6}$$

$$\Leftrightarrow \frac{d}{dt}\int_{V^*(t)} \rho(\mathbf{x}, t) dV + \int_{A^*(t)} \rho(\mathbf{x}, t)\mathbf{u}(\mathbf{x}, t) \cdot \mathbf{n} dA - \int_{A^*(t)} \rho(\mathbf{x}, t)\mathbf{b} \cdot \mathbf{n} dA = 0 \tag{2.1.0.7}$$

$$\Leftrightarrow \frac{d}{dt}\int_{V^*(t)} \rho(\mathbf{x}, t) dV + \int_{A^*(t)} \rho(\mathbf{x}, t)(\mathbf{u}(\mathbf{x}, t) - \mathbf{b}) \cdot \mathbf{n} dA = 0 \tag{2.1.0.8}$$

Eq. (2.1.0.8) is the statement of mass conservation for an arbitrary control volume. Note that $\mathbf{u}$, the velocity of the material surface, and $\mathbf{b}$, the velocity of the control surface, must be observed in the same coordinate system (the same frame of reference).

2.2 The Continuity Equation

Let's dive deeper into the material volume mass conservation statement to derive a differential form of mass conservation. Recall Gauss' theorem, which states that *for any* C^1 *vector field* $\mathbf{F}$ *defined on an open and bounded subset* $V \subset \mathbb{R}^n$

$$\int_V \nabla \cdot \mathbf{F} dV = \int_A \mathbf{F} \cdot \mathbf{n} dA \tag{2.2.0.1}$$

Where $A = \partial V$ is the boundary of the volume V. Applying Gauss' theorem to the surface integral in eq. (2.1.0.3) gives

$$\int_{V(t)} \frac{\partial \rho(\mathbf{x}, t)}{\partial t} + \nabla \cdot (\rho(\mathbf{x}, t)\mathbf{u}(\mathbf{x}, t)) dV = 0 \tag{2.2.0.2}$$

Since this must be true for every volume, the integrand must vanish at every point in space, which finally leaves us with

$$\frac{\partial \rho(\mathbf{x}, t)}{\partial t} + \nabla \cdot (\rho(\mathbf{x}, t)\mathbf{u}(\mathbf{x}, t)) = 0 \tag{2.2.0.3}$$

The first term is the rate of change of the mass density, while the second term is the divergence of the mass density flux. The continuity equation can be re-written using the following product rule

$$\nabla \cdot (\rho \mathbf{u}) = \rho \nabla \cdot \mathbf{u} + (\mathbf{u} \cdot \nabla)\rho$$

And the definition of the material derivative (1.2.0.7)

$$\frac{D\rho}{Dt} = \frac{\partial \rho}{\partial t} + \mathbf{u} \cdot \nabla \rho$$

To obtain the so-called *continuity equation*

$$\frac{1}{\rho}\frac{D\rho}{Dt} + \nabla \cdot \mathbf{u} = 0 \tag{2.2.0.4}$$

Note that for *incompressible* flow the material derivative vanishes and the continuity equation becomes

$$\nabla \cdot \mathbf{u} = 0 \tag{2.2.0.5}$$

A flow is incompressible if the density of the fluid is constant throughout the flow. Eq. (2.2.0.5) is sometimes considered the definition of incompressibility, since it is equivalent to the condition that the density is constant.

2.3 Conservation of Momentum in an arbitrary Control Volume

Newton's second law applied to a material volume $V(t)$ looks like this

$$\frac{d}{dt}\int_{V(t)} \rho(\mathbf{x}, t)\mathbf{u}(\mathbf{x}, t) dV = \frac{d}{dt}\int_{V(t)} \rho(\mathbf{x}, t)\mathbf{g} dV + \frac{d}{dt}\int_{A(t)} \mathbf{f}(\mathbf{n}, \mathbf{x}, t) dA \tag{2.3.0.1}$$

Here $\rho\mathbf{u}$ is the momentum per unit volume of flowing fluid, $\mathbf{g}$ is the body force (body forces are forces acting inside the material volume) per unit mass acting on the fluid within V and $\mathbf{f}$ is the surface force per unit area acting on A. $\mathbf{n}$ is the outward normal vector on A. Applying Reynolds transport theorem (2.1.0.2) gives

$$\int_{V(t)} \frac{\partial}{\partial t}(\rho(\mathbf{x},t)\mathbf{u}(\mathbf{x},t))dV + \int_{A(t)} \rho(\mathbf{x},t)\mathbf{u}(\mathbf{x},t)(\mathbf{u}(\mathbf{x},t)\cdot\mathbf{n})dA$$
$$= \int_{V(t)} \rho(\mathbf{x},t)\mathbf{g}dV + \int_{A(t)} \mathbf{f}(\mathbf{n},\mathbf{x},t)dA \tag{2.3.0.2}$$

The result for an arbitrary control volume is achieved through the same process as for conservation of mass, so I will just quote the result

$$\int_{V^*(t)} \frac{\partial}{\partial t}(\rho(\mathbf{x},t)\mathbf{u}(\mathbf{x},t))dV + \int_{A^*(t)} \rho(\mathbf{x},t)\mathbf{u}(\mathbf{x},t)(\mathbf{u}(\mathbf{x},t)-\mathbf{b})\cdot\mathbf{n}dA$$
$$= \frac{d}{dt}\int_{V^*(t)} \rho(\mathbf{x},t)\mathbf{g}dV + \int_{A^*(t)} \mathbf{f}(\mathbf{n},\mathbf{x},t)dA \tag{2.3.0.3}$$

2.4 Tensor Representation of Body and Surface Forces

Body forces are forces that act internally within the volume V without physical contact. Examples are gravity and electromagnetic forces. Body forces are always distributed throughout the volume and proportional to the total mass (or charge, current, etc.) contained within the volume. Consider the case where the body forces are conservative. In this case, they may be written as the gradient of a potential function Φ

$$\mathbf{g} = -\nabla\Phi, \quad g_i = -\partial_i\Phi \tag{2.4.0.1}$$

The surface forces in the various coordinate directions may be expressed using the stress tensor T_{ij} and surface normal $\mathbf{n}$

$$\mathbf{f} = \mathbf{n}\cdot\mathbf{T}, \quad f_i = T_{ji}n_j \tag{2.4.0.2}$$

It can be shown that the stress tensor is symmetric using Cauchy's tetrahedron argument (add this to the mathematics section if you like suffering).

2.5 Differential form of Momentum Conservation (Cauchy's Equation)

To obtain a differential form of the momentum conservation equation for a material volume, we apply Gauss' theorem on the surface integrals in eq. (2.3.0.3)

$$\int_{A(t)} \rho(\mathbf{x},t)\mathbf{u}(\mathbf{x},t)(\mathbf{u}(\mathbf{x},t)-\mathbf{b})\cdot\mathbf{n}dA = \int_{V(t)} \nabla\cdot(\rho(\mathbf{x},t)\mathbf{u}(\mathbf{x},t))\mathbf{u}(\mathbf{x},t)dV = \int_{V(t)} \partial_i(\rho u_i u_j)dV \tag{2.5.0.1}$$

$$\int_{A(t)} \mathbf{f}(\mathbf{n},\mathbf{x},t)dA = \int_{A(t)} \mathbf{T}\cdot\mathbf{n}dA = \int_{V(t)} \nabla\cdot\mathbf{T}dV = \int_{V(t)} \partial_i T_{ij}dV \tag{2.5.0.2}$$

Substituting this into (2.3.0.3) and collecting the terms in one volume integral gives:

$$\int_{V(t)} \partial_t(\rho u_j) + \partial_i(\rho u_i u_j) - \rho g_j - \partial_i T_{ij} dV = 0 \tag{2.5.0.3}$$

Since this must hold for any volume, the integrand must be zero everywhere

$$\partial_t(\rho u_j) + \partial_i(\rho u_i u_j) = \rho g_j + \partial_i T_{ij} \tag{2.5.0.4}$$

We can rewrite this even further by expanding the :HS

$$\partial_t(\rho u_j) + \partial_i(\rho u_i u_j) = \rho \partial_t u_j + u_j \partial_t \rho + u_j \partial_i(\rho u_i) + \rho u_i \partial_i u_j \tag{2.5.0.5}$$

$$= \rho \partial_t u_j + u_j \left[\partial_t + \partial_i(\rho u_i)\right] + \rho u_i \partial_i u_j \tag{2.5.0.6}$$

The contents of the square bracket are zero, because they are the continuity equation

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = \partial_t + \partial_i(\rho u_i)$$

Which leaves us with

$$\partial_t(\rho u_j) + \partial_i(\rho u_i u_j) = \rho \partial_t u_j + \rho u_i \partial_i u_j \tag{2.5.0.7}$$

$$= \rho \frac{Du_j}{Dt} \tag{2.5.0.8}$$

So the final result (which is called Cauchy's equation of motion) is

$$\rho \frac{Du_j}{Dt} = \rho g_j + \partial_i T_{ij} \tag{2.5.0.9}$$

2.6 Constitutive Equation for a Newtonian Fluid

A *constitutive equation* for a fluid is an equation that relates stress and deformation in the fluid. To derive the constitutive equation for a Newtonian Fluid (the definition of which will be presented later) we must first express the stress tensor in a form that depends on the strain rate tensor.

2.6.1 Stress Tensor Interlude

Consider a fluid volume at rest. Since the volume is at rest, the stress forces cannot depend on the orientation of the volume (otherwise they could introduce a rotation). The stress must be isotropic and can only be normal to the surface of the material volume. Hence, the stress tensor can be written in the form

$$T_{ij} = -p\delta_{ij} \tag{2.6.1.1}$$

Here p is the thermodynamic pressure. The negative sign is chosen by convention such that the components of $\mathbf{T}$ are positive if they indicate tension, and negative if they indicate compression.

If the fluid begins to undergo motion, it will no longer be at rest and T_{ij} will develop additional dynamic components τ_{ij} because of the viscosity (internal friction) of the fluid. So for a fluid in motion

$$T_{ij} = -p\delta_{ij} + \tau_{ij} \tag{2.6.1.2}$$

This reduces to (2.6.1.1) when the fluid is no longer in motion. The contribution τ_{ij} is called the *deviatoric/viscous stress tensor*. Galilean invariance requires τ_{ij} to be independent of the absolute fluid velocity, which means that it can only depend on the velocity gradient $\nabla\mathbf{u}$. Furthermore, the deviatoric stress tensor can only depend on the symmetric parts of $\nabla\mathbf{u}$ since (by definition!) stresses only develop exist when fluid elements change shape. Hence, τ_{ij} can only depend on the strain rate tensor S_{ij}. We want τ_{ij} to depend linearly on S_{ij} with the condition $S_{ij} = 0 \implies \tau_{ij} = 0$, which is achieved by

$$\tau_{ij} = K_{ijmn} S_{mn} \tag{2.6.1.3}$$

Here K_{ijmn} is some fourth-order tensor that depends on the local thermodynamic state of the fluid and specifies how τ_{ij} depends on each component of S_{ij}. If the fluid medium is isotropic, the relationship between T_{ij} and S_{ij} should be isotropic too, which is only possible if K_{ijmn} is isotropic! This requirement allows us to write K_{ijmn} in the following form

$$K_{ijmn} = \lambda\delta_{ij}\delta_{mn} + \mu\delta_{im}\delta_{jn} + \gamma\delta_{in}\delta_{jm} \tag{2.6.1.4}$$

For some set of scalars λ, mu and γ. These scalars are the source of the dependence of K_{ijmn} on the thermodynamic state of the fluid.

We can go even further: Since T_{ij} is symmetric, τ_{ij} must be symmetric as well. And since S_{ij} is symmetric, this means that the fourth-order tensor K_{ijmn} must be symmetric in i and j, which can only be achieved if

$$\gamma = \mu \tag{2.6.1.5}$$

This leaves us with a much more tractable expression for the deviatoric stress tensor

$$\tau_{ij} = 2\mu S_{ij} + \lambda S_{mm}\delta_{ij} \tag{2.6.1.6}$$

Hence the stress tensor becomes

$$T_{ij} = -p\delta_{ij} + 2\mu S_{ij} + \lambda S_{mm}\delta_{ij} \tag{2.6.1.7}$$

Fig. 3 shows how the magnitude of the shear stress τ in a fluid depends on the velocity gradient. It has been shown experimentally that τ is approximately proportional to the velocity gradient, and that the proportionality factor is the *dynamic viscosity* μ

$$\tau = \mu\nabla\mathbf{u} \tag{2.6.1.8}$$

This is the definition of a *Newtonian Fluid*. We can gain more insight by considering the sum T_{ii}

$$T_{ii} = -3p + 2\mu S_{ii} + 3\lambda S_{mm} \tag{2.6.1.9}$$

$$= -3p + (2\mu + 3\lambda)\nabla\cdot\mathbf{u} \tag{2.6.1.10}$$

This implies that

$$p = -\frac{1}{3}T_{ii} + \left(\frac{2}{3}\mu + \lambda\right)\nabla\cdot\mathbf{u} \tag{2.6.1.11}$$

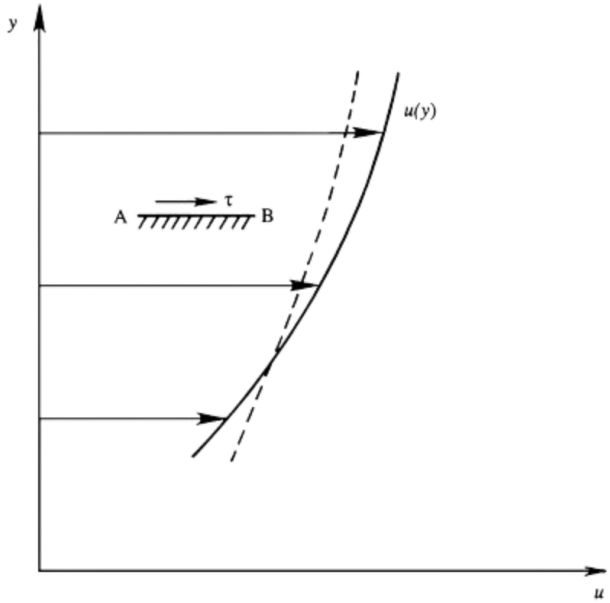


Figure 3: This illustration shows how the shear stress τ in a fluid is approximately proportional to the velocity gradient.

Now we define the mean pressure as the pressure in absence of shear and stress forces $\bar{p} = -\frac{1}{3}T_{ii}$ and substitute this into (2.6.1.11) to obtain

$$p - \bar{p} = \mu_v \nabla \cdot \mathbf{u} \tag{2.6.1.12}$$

Where we've introduced the **coefficient of bulk viscosity** $\mu_v = \frac{2}{3}\mu + \lambda$. *Stokes assumption* assumes that $\mu_v = 0$ such that $p = \bar{p}$. This is often a good assumption, since μ_v or $\nabla \cdot \mathbf{u}$ tend to be small for most flows.

2.6.2 The Constitutive Equation

Without Stokes assumption, the stress tensor is

$$T_{ij} = -p\delta_{ij} + \tau_{ij} \tag{2.6.2.1}$$

$$= -p\delta_{ij} + 2\mu\left(S_{ij} - \frac{1}{3}S_{mm}\delta_{ij}\right) + \mu_v S_{mm}\delta)ij \tag{2.6.2.2}$$

This is the constitutive equation for a Newtonian fluid. If the fluid is incompressible, $S_{mm} = \nabla \cdot \mathbf{u} = 0$, so the constitutive equation reduces to

$$T_{ij} = -p\delta_{ij} + 2\mu S_{ij} \tag{2.6.2.3}$$

2.6.3 Non-Newtonian fluid

Add the constitutive equation for a non-Newtonian fluid here.

2.7 The Navier-Stokes Momentum equation

To obtain a momentum-conservation equation for a Newtonian fluid, we substitute the constitutive equation (2.6.2.2) into Cauchy's equation of motion (2.5.0.9)

$$\rho \frac{Du_j}{Dt} = \rho g_j + \partial_i T_{ij} \tag{2.7.0.1}$$

$$= \rho g_j - \partial_i p + \partial_i \left[2\mu S_{ij} - \frac{2}{3}\mu S_{mm}\delta_{ij} + \mu_v S_{mm}\delta_{ij} \right] \tag{2.7.0.2}$$

$$= \rho g_j - \partial_i p + \partial_i \left[\mu(\partial_i u_j + \partial_j u_i) + \left(\mu_v - \frac{2}{3}\mu \right) \partial_m u_m \delta_{ij} \right] \tag{2.7.0.3}$$

Eq. (2.7.0.3) is called the *Navier-Stokes momentum equation* or simply the Navier-Stokes equation. When the temperature differences are small within the flow, μ and μ_v can be regarded as locally constant, which allows us to write

$$\rho \frac{Du_j}{Dt} = \rho g_j - \partial_i p + \mu \partial_i^2 u_j + \mu \partial_j \partial_i u_i + \left(\mu_v - \frac{2}{3}\mu \right) \partial_i \partial_m u_m \delta_{ij} \tag{2.7.0.4}$$

$$= \rho g_j - \partial_i p + \mu \partial_i^2 u_j + \left(\mu_v + \frac{1}{3}\mu \right) \partial_j \partial_m u_m \tag{2.7.0.5}$$

This equation applies to compressible flow. For incompressible fluids $S_{mm} = 0$, so

$$\rho \frac{Du_j}{Dt} = \rho g_j - \partial_i p + \mu \partial_i^2 u_j \tag{2.7.0.6}$$

Or in vector notation

$$\boxed{\rho \frac{D\mathbf{u}}{Dt} = \rho \mathbf{g} - \nabla p + \mu \nabla^2 \mathbf{u}} \tag{2.7.0.7}$$

This is the Navier-Stokes equation for *incompressible, Newtonian fluids.*

Note that the viscous term is related to the vorticity ω in the following way

$$\mu \nabla^2 \mathbf{u} = -\mu \nabla \times \omega \tag{2.7.0.8}$$

For inviscid flows, this term drops out. The resulting Navier-Stokes equation is called the *Euler equation*

$$\rho \frac{D\mathbf{u}}{Dt} = \rho \mathbf{g} - \nabla p \tag{2.7.0.9}$$

3 Special Forms of the Equations

3.1 Navier-Stokes in a non-inertial frame of reference

The continuity equation (2.2.0.4) and the Navier-Stokes equation (2.7.0.7) were derived in an inertial frame of reference. In a non-inertial frame of reference, the continuity equation is unchanged, but the Navier-Stokes equation must be modified. Consider a frame of reference O' that undergoes translational motion with the velocity $\mathbf{U}(t)$ and rotational motion with the angular velocity $\Omega(t)$ with respect to the frame O. A fluid particle located at $\mathbf{x}' = (x_1', x_2', x_3')$ in O' will be located at the following position in O

$$\mathbf{x} = \mathbf{X} + \mathbf{x}'$$

Where $\mathbf{X}(t)$ is the position of the origin of O' as seen from O. The velocity of a fluid particle is then

$$\begin{aligned}
\mathbf{u} &= \frac{d\mathbf{x}}{dt} \\
&= \frac{d\mathbf{X}}{dt} + \frac{d\mathbf{x}'}{dt} \\
&= \mathbf{U} + \frac{d}{dt}\left(x_1'\mathbf{e}_1 + x_2'\mathbf{e}_2 + x_3'\mathbf{e}_3\right) \\
&= \mathbf{U} + \frac{dx_1'}{dt}\mathbf{e}_1 + \frac{dx_2'}{dt}\mathbf{e}_2 + \frac{dx_3'}{dt}\mathbf{e}_3 + \underbrace{x_1'\frac{d\mathbf{e}_1}{dt} + x_2'\frac{d\mathbf{e}_2}{dt} + x_3'\frac{d\mathbf{e}_3}{dt}}_{=\Omega\times\mathbf{x}'}
\end{aligned}$$

So we end up with

$$\mathbf{u} = \mathbf{U} + \mathbf{u}' + \Omega \times \mathbf{x}' \tag{3.1.0.1}$$

We proceed to find the acceleration

$$\begin{aligned}
\mathbf{a} &= \frac{d\mathbf{u}}{dt} \\
&= \frac{d\mathbf{U}}{dt} + \frac{d\mathbf{u}'}{dt} + \frac{d}{dt}(\Omega \times \mathbf{x}') \\
&= \frac{d\mathbf{U}}{dt} + \underbrace{\frac{du_1'}{dt}\mathbf{e}_1 + \frac{du_2'}{dt}\mathbf{e}_2 + \frac{du_3'}{dt}\mathbf{e}_3}_{\mathbf{a}'} + \underbrace{u_1'\frac{d\mathbf{e}_1}{dt} + u_2'\frac{d\mathbf{e}_2}{dt} + u_3'\frac{d\mathbf{e}_3}{dt}}_{\Omega\times\mathbf{u}'} + \frac{d\Omega}{dt} \times \mathbf{x}' + \underbrace{\Omega \times \frac{d\mathbf{x}'}{dt}}_{\Omega\times(\mathbf{u}'+\Omega\times\mathbf{x}')}
\end{aligned}$$

Each term in the expression has a physical interpretation

$$\mathbf{a} = \underbrace{\frac{d\mathbf{U}}{dt}}_{\text{translational acceleration of } O'} + \underbrace{\mathbf{a}'}_{\text{particle acceleration in } O'} + \underbrace{2\Omega \times \mathbf{u}'}_{\text{Coriolis acceleration}} \tag{3.1.0.2}$$

$$+ \underbrace{\frac{d\Omega}{dt} \times \mathbf{x}'}_{\text{angular acceleration of } O'} + \underbrace{\Omega \times (\Omega \times \mathbf{x}')}_{\text{centripital acceleration}} \tag{3.1.0.3}$$

Substituting $\frac{D\mathbf{u}}{dt} = \mathbf{a}(t)$ into the Navier-Stokes equation (2.7.0.7) leads to the non-inertial form of the equation

$$\rho\frac{D'\mathbf{u}'}{Dt} = -\nabla' p + \rho\left[\mathbf{g} - \frac{d\mathbf{U}}{dt} - 2\Omega\times\mathbf{u}' - \frac{d\Omega}{dt}\times\mathbf{x}' - \Omega\times(\Omega\times\mathbf{x}')\right] + \mu\nabla^{2'}\mathbf{u}' \quad (3.1.0.4)$$

Note that the presence of a prime sign indicates whether the differentiation takes place with respect to the coordinates of O' or O. The physical interpretation of the above equation is that the non-inertial frame corresponds to the addition of multiple new body force terms.

3.2 Navier-Stokes in (non-inertial) spherical coordinates

It is possible to express (3.1.0.4) in spherical coordinates, which is occasionally useful, since the Earth is nearly spherical. The calculation involves a lot of simple but very tedious vector calculus, so I will just state the result here. To perform the calculation yourself, simply consider the spherical coordinate expressions for the gradient, Laplacian, cross product and so on, as well as the derivatives of the spherical coordinate unit vectors. Anyhow, in the spherical coordinates shown in fig. 4, the non-inertial Navier-Stokes equation becomes

$$\begin{aligned}\frac{Du}{Dt} - \left(2\Omega + \frac{u}{r\cos\phi}\right)(v\sin\phi - w\cos\phi) &= -\frac{1}{\rho r\cos\theta}\partial_\phi p\\ \frac{Dv}{Dt} + \frac{wv}{r} + \left(2\Omega + \frac{u}{r\cos\theta}\right)u\sin\theta &= -\frac{1}{\rho r}\partial_\phi p\\ \frac{Dw}{Dt} - \frac{u^2+v^2}{r} - 2\Omega u\cos phi &= -\frac{1}{\rho}\partial_r p - g\end{aligned} \quad (3.2.0.1)$$

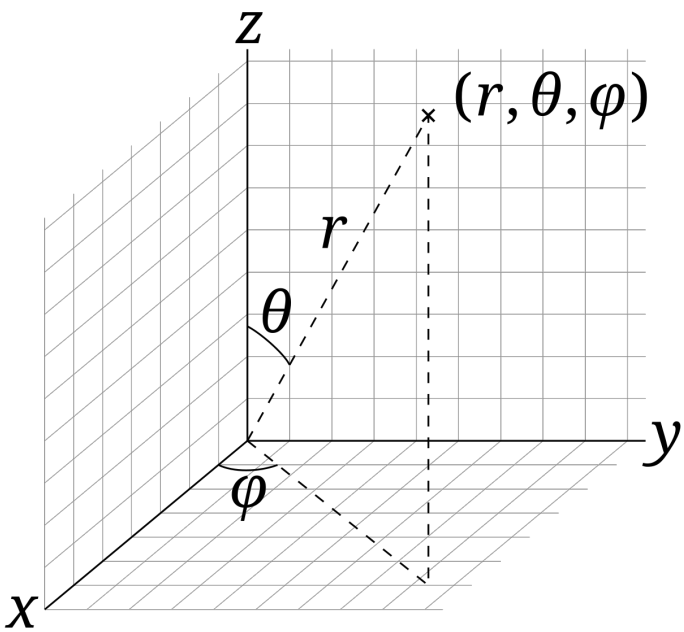


Figure 4: An illustration of spherical coordinates.

3.3 The Boussinesq approximation

At small velocities, the incompressible Navier-Stokes equation (2.7.0.7) becomes

$$0 = -\nabla p_s + \rho_s \mathbf{g} \tag{3.3.0.1}$$

Where p_s is the hydrostatic pressure of the fluid medium and ρ_s is the hydrostatic density. In this context, "hydrostatic" refers to the values of the quantity when the system is in a hydrostatic equilibrium. Subtract this from (2.7.0.7) to get

$$\rho \frac{D\mathbf{u}}{dt} = -\nabla p' + \rho' \mathbf{g} + \mu \nabla^2 \mathbf{u} \tag{3.3.0.2}$$

Where the prime denotes the difference from the hydrostatic value, i.e. $p' = p - p_s$ and $\rho' = \rho - \rho_s$. Eq. (3.3.0.2) is referred to as the Boussinesq approximation. It describes how small perturbations of p and ρ from equilibrium affects the fluid. It is only valid if the system is close to equilibrium and all variations are small, i.e. $\delta T \ll 1$, $\rho' \ll \rho_0$ and so on.

3.4 The theory of ideal fluids

Much like thermodynamics has a theory of ideal gasses, fluid mechanics has a theory of ideal fluids. The flow of an ideal fluid is governed by three equations:

The the continuity equation

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = 0 \tag{3.4.0.1}$$

Euler's equation of motion

$$\frac{D\mathbf{u}}{Dt} = -\frac{1}{\rho} \nabla p + \mathbf{f} \tag{3.4.0.2}$$

Here, $\mathbf{f}$ represents external forces.

The final equation is the equation for the adiabatic motion of a fluid particle

$$\partial_t s + \mathbf{u} \cdot \nabla s = 0 \tag{3.4.0.3}$$

Where s is the entropy per unit mass of the fluid.

3.5 Bernoulli's theorem

Consider an ideal fluid with uniform entropy, constant entropy per unit mass s and where the external force is conservative $\mathbf{f} = -\nabla \phi$. In that case, it is possible to show that Euler's equation of motion becomes

$$\boxed{\partial_t \mathbf{u} + \omega \times \mathbf{u} = -\nabla \left(\frac{|\mathbf{u}|^2}{2} + h + \phi \right)} \tag{3.5.0.1}$$

Where ω is the vorticity and $h = e + p/\rho$ is the enthalpy per unit mass, with e being the internal energy. Two very useful equations can be derived from this equation. The

first one can be found by considering irrotational flows, while the other can be found by considering steady flows.

For steady flows, all time derivatives vanish because the field variables are independent of time, in which case we get

$$\omega \times \mathbf{u} = -\nabla \left(\frac{|\mathbf{u}|^2}{2} + h + \phi \right) \tag{3.5.0.2}$$

Taking the inner product with $\mathbf{u}$ causes the left hand side to vanish, leading to

$$\mathbf{u} \cdot \nabla \left(\frac{|\mathbf{u}|^2}{2} + h + \phi \right) = \partial_\xi \left(\frac{|\mathbf{u}|^2}{2} + h + \phi \right) = 0 \tag{3.5.0.3}$$

The derivative is along a stream line parametrized by the parameter ξ. In other words, this means that the function H

$$\boxed{H \equiv \frac{|\mathbf{u}|^2}{2} + h + \phi} \tag{3.5.0.4}$$

is constant along streamlines for steady flows. This result is called the **Bernoulli theorem**. Surfaces of constant H are called **Bernoulli surfaces**.

Furthermore, if the inner product of ω and $\omega \times \mathbf{u}$ vanishes, H is analogously constant along vortex lines!

3.6 Bernoulli's equation

If the fluid density is constant everywhere $\rho = \rho_0$ the potential for the external forces becomes $\phi = gz$. Then applying the Bernoulli theorem (3.5.0.4) gives

$$H = \frac{|\mathbf{u}|^2}{2} + \frac{p}{\rho_0} + gz = \text{const.} \tag{3.6.0.1}$$

This is called **Bernoulli's equation**. The terms in the equation represent kinetic energy, pressure potential energy and gravitational potential energy (all per unit mass), so Bernoulli's equation is really a restatement of the conservation of energy for the fluid. The following conditions are needed to guarantee the validity of the equation:

- It can only be applied along a streamline, because the mechanical energy is constant on each streamline (although it can differ between streamlines).
- It can only be used for steady flow, since the mechanical energy of fluid elements is not conserved in an unsteady flow.
- It can only be used for inviscid flow, since viscosity leads to internal friction in the fluid, which allows mechanical energy to be converted into internal energy.
- It can only be used for incompressible flows, since the internal energy of a compressible flow changes as it is compressed or expanded (e.g. gasses heat up when compressed and cool down when expanded).

4 Boundary Conditions

The following sections include some examples of boundary conditions that can be used to solve the Navier-Stokes equation in various contexts.

4.1 Conservation of Mass Boundary Condition

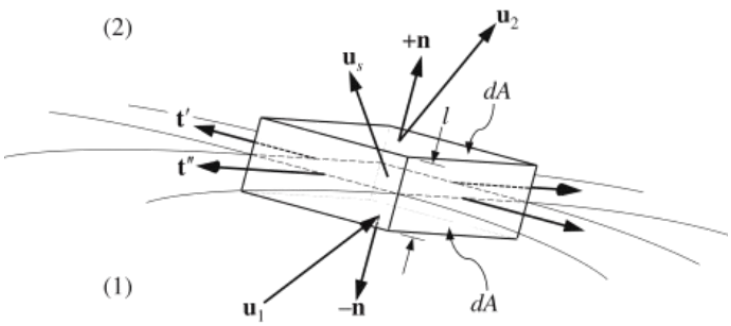


Figure 5: A rectangular control volume at the interface between two immiscible fluids (or a fluid and a solid). The control volume is moving with velocity $\mathbf{u}_s$ and the two tangent vectors to the interface surface are $\mathbf{t}'$ and $\mathbf{t}''$. $+\mathbf{n}$ and $-\mathbf{n}$ are the normal vectors pointing into medium 2 and medium 1 respectively. $\mathbf{u}_1$ and $\mathbf{u}_2$ are the velocities of the material surfaces in the two media.

Consider a control volume at the interace between two immiscible fluids as shown in fig. 5. We can apply eq. (2.1.0.8) for the conservation of mass in an arbitrary control volume to get

$$\dot{m}_s = \rho_1(\mathbf{u}_1 - \mathbf{u}_s) \cdot \mathbf{n} = \rho_2(\mathbf{u}_2 - \mathbf{u}_s) \cdot \mathbf{n} \tag{4.1.0.1}$$

Note that the sign convention chosen for $\dot{m}_s$ dictates whether $+\mathbf{n}$ or $-\mathbf{n}$ should be used for the normal vector. Some common cases are:

1. If (1) is immiscible with (2) there is no mass flowing across the boundary, hence $\dot{m}_s = 0$ and we get $\mathbf{u}_1 \cdot \mathbf{n} = \mathbf{u}_2 \cdot \mathbf{n}$.
2. If (1) is a solid that is not dissolving, ablating, sublimating or otherwise exchanging material at the interface, then $\dot{m}_s = 0$ and $\mathbf{u}_1 = \mathbf{u}_s$ so we get $\mathbf{u}_2 \cdot \mathbf{n} = \mathbf{u}_s \cdot \mathbf{n}$.
3. If the solid is also stationary, we get $\mathbf{u}_2 \cdot \mathbf{n} = 0$

4.2 Constant Surface Tension Boundary Condition

The boundary condition for surface tension can be derived from thermodynamic considerations. First recall that the Helmholtz free energy is defined as

$$F = U - TS \tag{4.2.0.1}$$

Where U is the internal energy of the system, T is the thermodynamical temperature and S is the entropy. To find the Helmholtz free energy per unit mass, we replace U and S by their per unit mass equivalents

$$f = e - Ts \tag{4.2.0.2}$$

Now consider an interface with area A that separates two fluids (1) and (2), with a surface tension coefficient σ. The surface tension coefficient corresponds to the free Helmholtz energy per unit area stored in the interface. Then the total free Helmholtz energy of the system is

$$F = \rho_1 V_1 f_1 + \rho_2 V_2 f_2 + \sigma A \tag{4.2.0.3}$$

The sign of σ is physically meaningful. Recall that nature tends to minimize the Helmholtz free energy.

- If $\sigma > 0$, the two media are immiscible and F will be at a local minimum when A is small, so the interface surface area will tend to decrease over time.
- If $\sigma < 0$, the two media are miscible and the minimum of F occurs when A is large, so the interface surface area will tend to increase until the fluids are completely mixed and the spacing between the folds of the interface is on the order of the molecular distances in the mixture.

Locally, the part of the interface surface that is bounded by some curve C can be seen as a half-spheroid with two principal radii of curvature R_1 and R_2, see fig. 6.

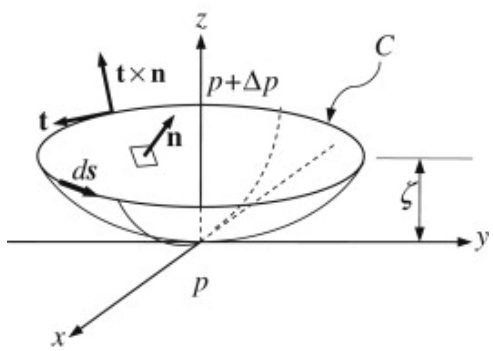


Figure 6: A local model of the interface surface.

We are interested in the equilibrium shape in the case where $\sigma > 0$. In this case, the shape will be dictated by the pressure difference Δp between the lower and upper region in the figure, and the shape will be described by some function η of the following form

$$\eta(x, y, z) = z - h(x, y) = 0 \tag{4.2.0.4}$$

Now choose the coordinate system such that $h(\mathbf{0}) = \partial_i h(\mathbf{0}) = 0$ and such that the x- and y-directions are the direction along which the principal axes curvature lie along the coordinate axes (meaning R_1 and R_2 are found in the xz- and yz-planes). In that case, the shape of the interface is given by

$$\eta(x, y, z) = z\left(\frac{x^2}{2R_1}\right) - \left(\frac{y^2}{2R_2}\right) = 0 \tag{4.2.0.5}$$

4.2.1 Balance of Forces Calculation

If we want to understand how Δp relates to the curvature R_1 and R_2 when the pressure forces are balanced, we will need to perform some detailed calculations. We will consider the limit where the interface area enclosed by C approaches 0.

First, we need to determine the net pressure force $\mathbf{F}_p$ on the interface surface A. The pressure force can be computed via the following surface integral

$$\mathbf{F}_p = -\int_A \Delta p \mathbf{n} dA \tag{4.2.1.1}$$

The minus sign is because we have chosen $\mathbf{n}$ to be the outward normal, and we want a higher pressure above the interface ($\Delta p > 0$) to lead to a downwards pressure force. See fig. 6.

Finding $\mathbf{n}$ is simple

$$\mathbf{n} = \frac{\nabla \eta}{|\nabla \eta|} = \frac{(-x/R_1, -y/R_2, 1)}{\sqrt{(x/R_1)^2 + (y/R_2)^2 + 1}} \tag{4.2.1.2}$$

The area element is

$$dA = \sqrt{1 + (\partial_x \eta)^2 + (\partial_y \eta)^2} dxdy = \sqrt{1 + (x/R_1)^2 + (y/R_2)^2} dxdy \tag{4.2.1.3}$$

We are now ready to perform the integral. In equilibrium, Δp is constant and can be taken outside of the integral

$$\mathbf{F}_p = -\Delta p \int_{-\sqrt{2R_1\xi}}^{+\sqrt{2R_1\xi}} \left[\int_{-\sqrt{2R_2\xi - x^2 R_2/R_1}}^{+\sqrt{2R_2\xi - x^2 R_2/R_1}} (-x/R_1, -y/R_2, 1) dy \right] dx \tag{4.2.1.4}$$

The limits of integration arise from the parametrization of the half-spheroid. The z-component can be evaluated relatively easily

$$(\mathbf{F}_p)_z = -\pi \Delta p \sqrt{2R_1\xi} \sqrt{2R_2\xi} \tag{4.2.1.5}$$

The integrals for the other components are odd integrals with symmetrical limits, so they are zero.

The surface tension force $\mathbf{F}_{st}$ acting on the curve C can be found with the line integral

$$\mathbf{F}_{st} = \sigma \oint_C \mathbf{t} \times \mathbf{n} ds \tag{4.2.1.6}$$

The tangent vector $\mathbf{t}$ to C is

$$\mathbf{t} = -\frac{(1, dy/dx, 0)}{\sqrt{1 + (dy/dx)^2}} \tag{4.2.1.7}$$

And the line element is

$$ds = \sqrt{1 + (dy/dx)^2} dx \tag{4.2.1.8}$$

By performing the cross product and inserting the expression for ds, it is possible to show that

$$\mathbf{t} \times \mathbf{n} ds = \frac{R_2/y)}{sqrt1 + (dy/dx)^2} \left(\frac{x}{R_1}, \frac{y}{R_2}, \frac{x^2}{R_1^2} + \frac{y^2}{R_2^2} \right) dx \tag{4.2.1.9}$$

$$\approx \frac{R_2}{y} \left(\frac{x}{R_1}, \frac{y}{R_2}, \frac{x^2}{R_1^2} + \frac{y^2}{R_2^2} \right) dx \tag{4.2.1.10}$$

The approximation holds for $x/R_1, y/R_2 \ll 1$. We can eliminate y using the equation for the boundary curve

$$\xi = (x^2/2R_1) - (y^2/2R_2), \quad (\xi = \text{const}) \tag{4.2.1.11}$$

Doing so leads to the following integral for the z-component (the x- and y-components are both zero due to the symmetry of the integration path)

$$(\mathbf{F}_{st})_z = 4\sigma \int_0^{\sqrt{2R-1\xi}} \frac{R-2}{\sqrt{2R_2\xi - (R_2/R_1)x^2}} \left[2\xi + \frac{x^2}{R_1}\left(\frac{1}{R_1} - \frac{1}{R_2}\right)\right] dx \tag{4.2.1.12}$$

The factor of 4 is because the integral with these limits only covers a quarter of the full integration path shown in fig. 6. Performing this integral gives

$$(\mathbf{F}_{st})_z = \pi\sigma\sqrt{2R_1\xi}\sqrt{2R_2\xi}\left(\frac{1}{R_1} + \frac{1}{R_2}\right) \tag{4.2.1.13}$$

We are almost done now. Pressure equilibrium requires the net pressure force and net surface tension force to cancel

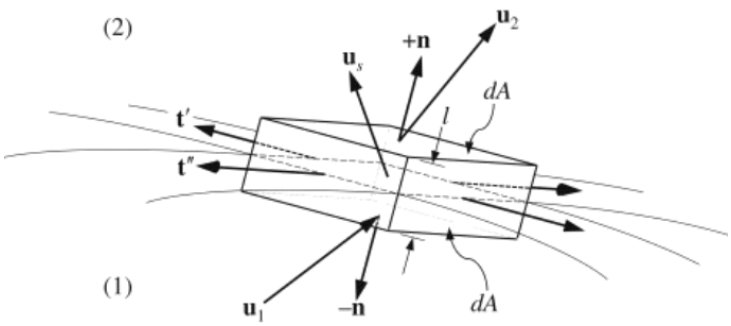
$$\mathbf{F}_p + \mathbf{F}_{st} = 0 \tag{4.2.1.14}$$

Which implies that

$$\Delta p = \sigma\left(\frac{1}{R_1} + \frac{1}{R_2}\right) \tag{4.2.1.15}$$

4.3 Conservation of Momentum Boundary Condition

Recall fig. 5.



If we write down the conservation of momentum along the normal direction $\mathbf{n}$ and use eq. (4.2.1.15), we get

$$\dot{m}_s(\mathbf{u}_2 - \mathbf{u}_1)\cdot\mathbf{n} = -\Delta p + ((n_i\tau_{ij})_2 - (n_i\tau_{ij})_1) + \sigma\left(\frac{1}{R'} + \frac{1}{R''}\right) \tag{4.3.0.1}$$

Where τ_{ij} is the viscous stress tensor (2.6.1.6) and R' and R'' are the principal radii of curvature that lie along the principal directions of curvature $\mathbf{t}'$ and $\mathbf{t}''$. When $\dot{m}_s = 0$ and the fluid is inviscid, or when both fluids are stationary, this reduces to the pressure differnece (4.2.1.15).

4.3.1 The No-Slip Condition

It is not possible to ascertain the requirements on the tangential components of the fluid velocities from the equations of motion. However, the following condition has been experimentally verified and is accepted for most ordinary fluids

$$\mathbf{u}_1 \cdot \mathbf{t}' = \mathbf{u}_2 \cdot \mathbf{t}', \quad \mathbf{u}_1 \cdot \mathbf{t}'' = \mathbf{u}_2 \cdot \mathbf{t}'' \tag{4.3.1.1}$$

This is called the *no-slip condition* and it states that the tangential velocity components must match at the interface. For the common case of a viscous fluid (2) moving with respect to an impermeable solid (1), eqs. (4.3.0.1) and (4.3.1.1) reduce to

$$\mathbf{u}_2 = \mathbf{u}_1 \tag{4.3.1.2}$$

At the interface.

4.3.2 Control Volume Boundary Conditions

For the control volume in fig. 5, the following boundary conditions apply

$$((n_i\tau_{ij})_2 - (n_i\tau_{ij})_1)t'_j + (\partial_j\sigma)t'_j = ((n_i\tau_{ij})_2 - (n_i\tau_{ij})_1)t''_j + (\partial_j\sigma)t''_j = 0 \tag{4.3.2.1}$$

4.4 Conservation of Energy Boundary Conditions

The conservation of energy boundary condition can be found by considering the internal and kinetic energy of the fluids, the work done by body and surface forces, as well as the heat flux across the interface. By accounting for these factors, it is possible to arrive at

$$\dot{m}_s \left[\left(h + \frac{1}{2}|\mathbf{u}|^2 \right)_2 - \left(h + \frac{1}{2}|\mathbf{u}|^2 \right)_1 \right] = -\kappa \left[(\nabla T)_2 - (\nabla T)_1 \right] \cdot \mathbf{n} \tag{4.4.0.1}$$

Where h is the heat transfer coefficient and κ is the thermal conductivity across the interface. The energy equivalent of the no-slip condition is

$$T_1 = T_2 \tag{4.4.0.2}$$

Which forbids temperature jumps at the boundary.

5 Introduction to Waves

5.1 Basic concepts

The following is a very general definition of a wave: *A wave is a dynamically propagating disturbance in a medium, one that typically propagates at a rate distinct from the motion of the medium.*

Waves are examples of solutions to the Navier-Stokes equation. If the system is locally homogeneous and the amplitude of the wave is small, the solutions are often **plane waves**. Let $\phi(\mathbf{x}, t)$ be some scalar field variable, such as pressure or temperature. A plane wave with amplitude A, wave vector $\mathbf{k}$ and frequency ω produces a dependence of the following form

$$\phi(\mathbf{x}, t) = \mathrm{Re}\left(Ae^{i(\mathbf{k}\cdot\mathbf{x}-\omega t}\right) \tag{5.1.0.1}$$

It is sometimes useful to introduce the **phase** of the wave $\theta = \mathbf{k}\cdot\mathbf{x} - \omega t$. Note that a plane wave is spatially periodic with a wavelength λ. Consider a scalar distance s perpendicular to a line of constant phase. By considering this situation, we see that the wavelength related to the wave number $|\mathbf{k}|$ by

$$\begin{aligned} \phi(ks) &= \phi(ks + \lambda) \\ \implies |\mathbf{k}| &= \frac{2\pi}{\lambda} \end{aligned} \tag{5.1.0.2}$$

Thus the wavelength λ is the distance between the two nearest points where the wave has the same phase. It is also worth noting that

$$\frac{d\theta}{dt} = -\omega \tag{5.1.0.3}$$

So ω is the rate of decrease of the phase. There is also a temporal periodicity with period T related to ω by

$$T = \frac{2\pi}{\omega} \tag{5.1.0.4}$$

T is the shortest time it can take for the same phase to reappear.

The **phase velocity** of the wave is defined as the propagation velocity of the phase in the direction of the wave vector

$$\mathbf{c} = \frac{\omega}{k}\mathbf{e}_k \tag{5.1.0.5}$$

Note that the angular frequency and wave number are often related by a **dispersion relation** $\omega(k)$. The term dispersion relation stems from the fact that this leads wave components with different wave numbers to travel at different speeds, which causes wave packets to spread out or "disperse" over time.

The **group velocity** of the wave with a wave vector $\mathbf{k} = (k, l, m)$ is defined as

$$\mathbf{c}_g = (\partial_k\omega)\mathbf{e}_x + (\partial_l\omega)\mathbf{e}_y + (\partial_m\omega)\mathbf{e}_z \tag{5.1.0.6}$$

The group velocity is the velocity with which the overall wave envelope moves (as opposed to the individual wave fronts, which move with the phase velocity). The group velocity is also the velocity of propagation of the energy in the wave.

5.2 Example: Internal gravity waves in a continuously stratified fluid

Consider a continuously stratified fluid, with the following assumptions for simplicity

- The wave motion is inviscid because the velocity gradients are small and Reynolds number is large
- The wave amplitudes are small enough that nonlinear advection terms can be neglected
- The frequency of the wave motion is much larger than the Coriolis frequency, such that Coriolis effects are negligible
- The fluid is incompressible

These conditions, together with the Boussinesq approximation (3.3.0.2) produces the following equations

$$\underbrace{\frac{D\rho}{Dt} = 0}_{\text{constant fluid particle density}} \tag{5.2.0.1}$$

$$\underbrace{\nabla \cdot \mathbf{u} = 0}_{\text{incompressibilty}} \tag{5.2.0.2}$$

$$\partial_t u = -\frac{1}{\rho_0}\partial_x p' \tag{5.2.0.3}$$

$$\partial_t v = -\frac{1}{\rho_0}\partial_y p' \tag{5.2.0.4}$$

$$\partial_t w = -\frac{1}{\rho_0}\partial_z p' - \frac{\rho g}{\rho_0} \tag{5.2.0.5}$$

Where ρ_0 is the reference density of the fluid when no perturbations are present. Now define the quiescent density and pressure as those that satisfy the hydrostatic balance

$$0 = -\frac{1}{\rho_0}\partial_z \bar{p} 9z) - \frac{\bar{\rho}(z)g}{\rho_0} \tag{5.2.0.6}$$

This allows us to write the value of p and ρ as small perturbations from the quiescent values

$$p = \bar{p}(z) + p', \quad \rho = \bar{\rho}(z) + \rho' \tag{5.2.0.7}$$

The density equation (5.2.0.1) can be written as

$$\partial_t(\bar{\rho} + \rho') + u\partial_x(\bar{\rho} + \rho') + v\partial_y(\bar{\rho} + \rho') + w\partial_z(\bar{\rho} + \rho') = 0 \tag{5.2.0.8}$$

But the time derivative of $\bar{\rho}$ is zero, and the nonlinear terms containing ρ' vanish. Since $\bar{\rho}(z)$ is independent of x and y, this leaves just two terms

$$\boxed{\partial_t \rho' + w\partial_z \bar{\rho} = 0} \tag{5.2.0.9}$$

This is the linearized density equation. Physically, this means that the density perturbation at a point is only due to the vertical advection at that point. Using this equation as well as eqs. (5.2.0.2)-(5.2.0.5) gives

$$\partial_t u = -\frac{1}{\rho_0}\partial_x p' \tag{5.2.0.10}$$

$$\partial_t v = -\frac{1}{\rho_0}\partial_y p' \tag{5.2.0.11}$$

$$\partial_t w = -\frac{1}{\rho_0}\partial_z p' - \frac{\rho' g}{\rho_0} \tag{5.2.0.12}$$

$$\partial_t \rho' - \frac{N^2 \rho_0}{g} w = 0 \tag{5.2.0.13}$$

Where $N^2 = -\frac{g}{\rho_0}\partial_z \bar{\rho}$ is the Brunt-Väisälä frequency (also called the *buoyancy frequency*). We see that the perturbation variables p' and ρ' replace their equivalents when we linearize the system.

We can now take the time derivative of the incompressibility equation (5.2.0.2) and use the results we just found for $\partial_t u$ and $\partial_t v$ to get

$$\frac{1}{\rho_0}\nabla_H^2 p' = \partial_{zt} w \tag{5.2.0.14}$$

Where $\nabla_H^2 = \partial_{xx} + \partial_{yy}$ is the horizontal Laplacian operator. By applying the equation for $\partial_t w$ and $\partial_t \rho' - \frac{N^2\rho_0}{g} w = 0$, we get

$$\frac{1}{\rho_0}\partial_t p' = \partial_{zz} w - N^2 w \tag{5.2.0.15}$$

By combining this result with (5.2.0.14), it is possible to arrive at (SHOW THIS IN MORE DETAIL)

$$\partial_{tt}(\partial_{zz} w) = -\nabla_H^2 \left(\partial_{tt} w + N^2 w\right) \tag{5.2.0.16}$$

$$\Leftrightarrow \boxed{\partial_{tt}(\nabla^2 w) = -N^2 \nabla_H^2 w} \tag{5.2.0.17}$$

Eq. (5.2.0.17) is the governing equation for the vertical velocity w. It will be very useful when we proceed to derive the dispersion relation for internal gravity waves, which we will do in a moment.

Since the fluid is continuously stratified and the waves may propagate in any direction, the dispersion relation will be anisotropic

$$\omega = \omega(\mathbf{k}) \tag{5.2.0.18}$$

Where $\mathbf{k} = (k, l, m)$ is the wave vector. Consider a plane wave of the form

$$w = w_0 e^{i(\mathbf{k}\cdot\mathbf{x} - \omega t)} \tag{5.2.0.19}$$

Substituting this into the governing equation (5.2.0.17) (and assuming $N^2 = \text{const.}$) leads to the following dispersion relation

$$\omega^2 = \frac{k^2 + l^2}{k^2 + l^2 + m^2} N^2 \tag{5.2.0.20}$$

If we choose our coordinate system such that $\mathbf{k}$ lies in the xz-plane, this simplifies to

$$\omega = \frac{kN}{\sqrt{k^2 + m^2}}$$

The phase velocity and group velocity are

$$\mathbf{c} = \frac{\omega}{k}\mathbf{e}_k \tag{5.2.0.21}$$

$$= \frac{\omega}{|\mathbf{k}|^2}(k\mathbf{e} - x + m\mathbf{e}_z) \tag{5.2.0.22}$$

and

$$\mathbf{c}_g = (\partial_k \omega)\mathbf{e}_x + (\partial_l \omega)\mathbf{e}_y + (\partial_m \omega)\mathbf{e}_z \tag{5.2.0.23}$$

$$= \frac{Nm}{|\mathbf{k}|^3}(m\mathbf{e}_x - k\mathbf{e}_z) \tag{5.2.0.24}$$

Interestingly, the phase and group velocities are perpendicular. Their horizontal components point in the same direction, but their vertical components are opposite. Fig. 7 illustrates how this means that wave crests move side-to-side in a motion perpendicular to the wave envelope motion.

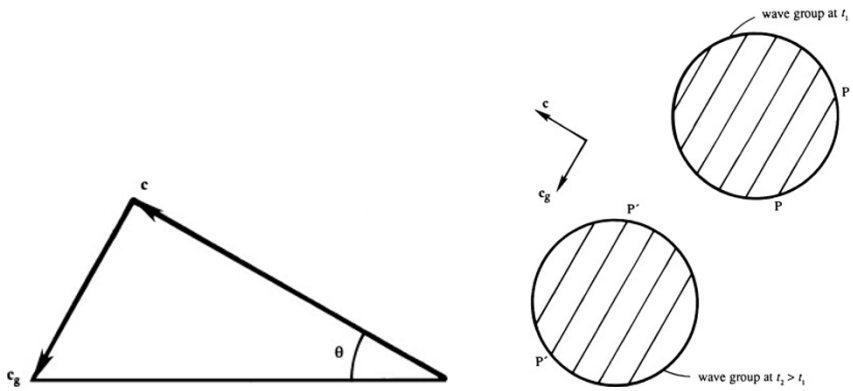


Figure 7: An illustration of the propagation of the internal gravity waves with perpendicular $\mathbf{c}$ and $\mathbf{c}_g$. Note how the wave crests move sideways compared to the motion of the wave envelopes.

6 Gravity Waves

6.1 Example: Surface gravity waves in a basin

In this section, we will discuss *gravity waves*, which are waves where gravity is the restoring force. We will assume that the wave frequency is much higher than the Coriolis frequency, such that we can neglect the Coriolis effect. We will also assume that the wave amplitudes are so small that all equations of motion and boundary conditions are linear. We will

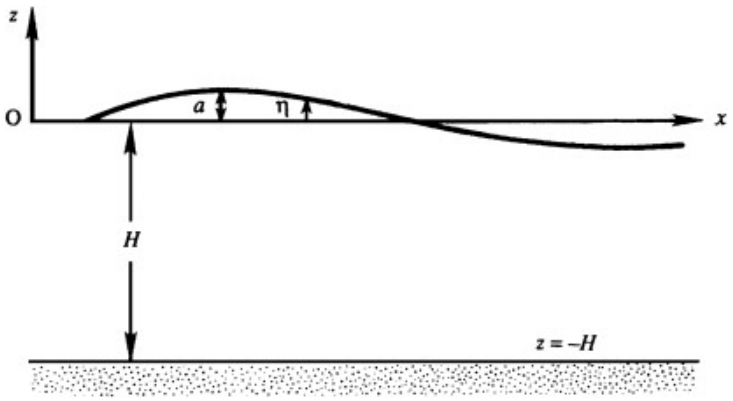


Figure 8: An illustration of the linear liquid-surface gravity-waves. The x-axis has been chosen in the direction of propagation of the wave. η measures the vertical deviation of the fluid surface from its equilibrium height $z = 0$ and H is the depth of the basin.

discuss waves of the kind shown in fig. 8. Our Ansatz will be that the waves are simple sine/cosine waves

$$\eta(x,t) = a\cos(kx - \omega t) \tag{6.1.0.1}$$

Remember that we are assuming $a/\lambda \ll 1$ and $a/H \ll 1$. The Ansatz is justified by the fact that long, low-amplitude surface gravity waves have often been observed to be sinusoidal.

Because the flow is irrotational, it is possible to define a velocity potential ϕ such that $\mathbf{u} = -\nabla\phi$. Assuming the flow is incompressible (which is a good assumption for water) means that $\nabla \cdot \mathbf{u} = 0$, which implies

$$\boxed{\nabla^2\phi = 0} \tag{6.1.0.2}$$

Now let's write down some boundary conditions. The first condition is that the fluid velocity at the bottom of the basin cannot be normal to the boundary

$$\boxed{w = \partial_z\phi = 0, \quad \text{at } z = -H} \tag{6.1.0.3}$$

The second condition is that fluid particles at the surface should move with the surface, which is expressed by stating that the normal component of the fluid velocity at the surface should be the same as the normal component of the surface velocity

$$(\mathbf{u}\cdot\mathbf{n})_{z=\eta} = \mathbf{u}_s \cdot \mathbf{n} \tag{6.1.0.4}$$

Let's try to write this in terms of the velocity potential. The equation describing the surface is $f(x, y, z) = z - \eta(x, t) = 0$, which tells us that the surface normal is

$$\mathbf{n} = \frac{\nabla f}{|\nabla f|} = \frac{(-\partial_x \eta, 0, 1)}{\sqrt{(\partial_x \eta)^2 + 1}} \tag{6.1.0.5}$$

The surface velocity is strictly vertical $\mathbf{u}_s = \partial_t \eta \mathbf{e}_z$. Hence

$$\begin{aligned} (\mathbf{u} \cdot \mathbf{n})_{z=\eta} &= \mathbf{u}_s \cdot \mathbf{n} \\ \Leftrightarrow (\mathbf{u} \cdot \nabla f)_{z=\eta} &= \mathbf{u}_s \cdot \nabla f \\ \Leftrightarrow \partial_z \phi\big|_{z=\eta} &= \partial_t \eta + (\partial_x \eta)(\partial_x \phi)\big|_{z=\eta} \end{aligned}$$

The slope of the wave is small because of the small amplitude assumption, which means that the term $(\partial_x \eta)(\partial_x \phi)\big|_{z=\eta}$ can be neglected. For consistency, the LHS must be Taylor expanded to first order as well, which gives

$$\boxed{\partial_z \phi\big|_{z=0} = \partial_t \eta} \tag{6.1.0.6}$$

This is the second boundary condition in terms of ϕ. It is valid if $a/\lambda \ll 1$.

A third boundary condition arises from one of the Bernoulli equations (SPECIFY)

$$\partial_t \phi + \frac{1}{2}|\nabla \phi|^2 + gz + \frac{p}{\rho} = \text{const.} \tag{6.1.0.7}$$

We can drop the non-linear term because of our approximations

$$\partial_t \phi + gz + \frac{p}{\rho} \approx 0 \tag{6.1.0.8}$$

The Bernoulli constant in this equation should be measured on the surface of the liquid, but far from the wave. To find the third boundary condition, we note that the pressure just below the surface should be the same as the ambient pressure, i.e. the Gauge pressure should be zero

$$p\big|_{z=0} = 0 \tag{6.1.0.9}$$

Evaluating the Bernoulli equation (6.1.0.8) at $z = \eta$ and Taylor expanding to first order produces the following result

$$\left(\partial_t \phi + gz + \frac{p}{\rho}\right)\Big|_{z=\eta} \approx \partial_t \phi\big|_{z=0} + g\eta \approx 0 \tag{6.1.0.10}$$

Which can be written as

$$\boxed{\phi\big|_{z=0} = -g\eta} \tag{6.1.0.11}$$

Time to solve the system to find ϕ. Since η is a cosine function of $kx - \omega t$, the boundary conditions require ϕ to be a sine function of $kx - \omega t$

$$\phi(\mathbf{x}, t) = f(z) \sin(kx - \omega(k)t) \tag{6.1.0.12}$$

Substitute this into the Laplace eq. (6.1.0.2)

$$\partial_z^2 f - k^2 f = 0 \tag{6.1.0.13}$$

This is a second-order, linear, homogeneous ODE for f with the general solution

$$f(z) = C_+ e^{kz} + C_- e^{-kz} \tag{6.1.0.14}$$

Substitute this back into the expression for ϕ to get

$$(C_+ e^{kz} + C_- e^{-kz}) \sin(kz - \omega(k)t) \tag{6.1.0.15}$$

One of the constants can be eliminated with (6.1.0.3)

$$k(C_+ e^{-kH} - C_- e^{kH}) \sin(kz - \omega(k)t) = 0 \implies C_- = C_+ e^{-2kH} \tag{6.1.0.16}$$

Now use (6.1.0.11) to get

$$\begin{aligned} k(C_+ - C_-) \sin(kz - \omega t) = \omega a \sin(kz - \omega t) &\implies k(C_+ - C_-) = \omega a \\ \Leftrightarrow kC_+(1 - e^{-2kH}) &= \omega a \\ \implies C_+ = \frac{\omega a}{k(1 - e^{-2kH})}, &\quad C_- = \frac{\omega a e^{-2kH}}{k(1 - e^{-2kH})} \end{aligned}$$

We are done now. Plugging this back into ϕ and rewriting the result at bit gives

$$\phi = \frac{\omega a}{k} \frac{\cosh(k(z + H)}{\sinh(kH)} \sin(kx - \omega t) \tag{6.1.0.17}$$

Which in turn gives the velocity field (v is zero)

$$\boxed{u = \omega a \frac{\cosh(k(z + H)}{\sinh(kH)} \cos(kx - \omega t), \quad w = \omega a \frac{\sinh(k(z + H)}{\sinh(kH)} \sin(kx - \omega t)} \tag{6.1.0.18}$$

We can also find the *dispersion relation*. Using (6.1.0.11) gives

$$-\frac{\omega^2}{k} \frac{\cosh(kH)}{\sinh(kH)} \cos(kx - \omega t) \approx -g\eta = -ga \cos(kx - \omega t) \tag{6.1.0.19}$$

$$\implies \boxed{\omega = \sqrt{gk \tanh(kH)}} \tag{6.1.0.20}$$

The associated phase speed is

$$\boxed{c = \frac{\omega}{k} = \sqrt{\frac{g}{k} \tanh(kH)}} \tag{6.1.0.21}$$

We see that ω depends on k. Waves for which this is true are called *dispersive* because the k-dependence causes waves with different wave numbers to travel at different speeds, which means that a wave packet made will not retain its shape as time progresses — the waves of different frequencies that make up the packet will *disperse* over time. This is why the formula for $\omega(k)$ is called the *dispersion relation*.

6.1.1 Deep Water Approximation

The deep water approximation is valid in the limit $H/\lambda \gg 1$. Note that $\tanh(x) \to 1$ for $x \to \infty$. This leads to the following dispersion relation and phase speed

$$\boxed{\omega = \sqrt{gk}, \quad c = \sqrt{\frac{g}{k}}} \tag{6.1.1.1}$$

The velocity field components in this limit are

$$\boxed{u = \omega a e^{kz} \cos(kx - \omega t), \quad w = \omega a e^{kz} \sin(kx - \omega t)} \tag{6.1.1.2}$$

6.1.2 Shallow Water Approximation

In the shallow water limit, $H/\lambda \ll 1$, we apply $\tanh(x) \approx x$ for small x to see that

$$\boxed{\omega = k\sqrt{gH}, \quad c = \sqrt{gH}} \tag{6.1.2.1}$$

The velocity field components in this limit are

$$\boxed{u = \frac{\omega a}{kH} \cos(kx - \omega t), \quad w = \omega a \left(1 + \frac{z}{H}\right) \sin(kx - \omega t)} \tag{6.1.2.2}$$

6.2 The energy associated with gravity waves

The potential energy E_p per unit area of a wave system is defined as the work required to deform a unit surface into the disturbed state it is in because of the wave. An equivalent definition defines E_p as the difference in the potential energy of the system in the disturbed and undisturbed states.

For a wave of the type shown in fig. 8, the potential energy element (per unit length along y) is $\rho g z dx dz$, so the potential energy can be computed as

$$E_p = \frac{\rho g}{\lambda} \int_0^\lambda \int_{-H}^{\eta} z dz dx - \frac{\rho g}{\lambda} \int_0^\lambda \int_{-H}^{0} z dz dx \tag{6.2.0.1}$$

$$= \frac{\rho g}{\lambda} \int_0^\lambda \int_0^{\eta} z dz dx = \frac{\rho g}{2\lambda} \int_0^\lambda \eta^2 dx \tag{6.2.0.2}$$

This can also be written in a more useful form involving the mean-square displacement

$$\boxed{E_p = \frac{1}{2} \rho g \overline{\eta^2}} \tag{6.2.0.3}$$

Incidentally, this means that the average kinetic and potential energies are equal. This is a fluid-mechanical *principle of equipartition of energy* that is valid when the displacement is small and Coriolis effects can be neglected. Hence the total wave energy in the water is

$$\boxed{E = E_p + E_k = \rho g \overline{\eta^2} = \frac{1}{2} \rho g a^2} \tag{6.2.0.4}$$

Where a is the amplitude of the wave if it is assumed to be of a sinusoidal form.

6.3 Gravity waves affected by the Coriolis force

In our previous discussion of surface gravity waves, we ignored the Coriolis force. In this section, we will see what happens when we include its effects.

6.3.1 The linearized shallow-water equations

Before we discuss gravity waves affected by the Coriolis force, we need to derive the linearized equations of motion for shallow-water waves.

Let's start by reminding ourselves that we are considering the same situation that we were in when we discussed surface gravity waves, see fig. 9. Note that we have chosen coordinates where $z = 0$ at the bottom of the basin. The fluid is assumed to be incompressible.

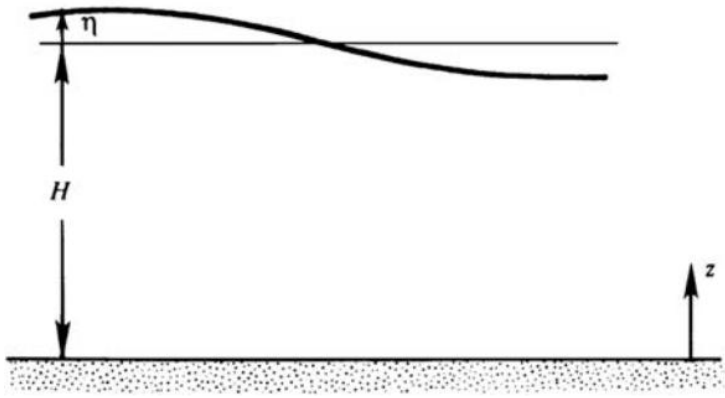


Figure 9: An illustration of the linear liquid-surface gravity-waves. The x-axis has been chosen in the direction of propagation of the wave. η measures the vertical deviation of the fluid surface from its equilibrium height $z = H$ and H is the depth of the basin.

Our goal will be to simplify the continuity equation and Navier-Stokes equation for this situation using the fact that the wave oscillations are small, which means that we can neglect any terms that are non-linear in the displacement η or velocity $\mathbf{u}$.

The pressure at any given point is $p = \rho g h$ where h is the height of the fluid column above that point. Since $h = (H+\eta)-z$ for a point (x, y, z), the pressure is given by

$$p(x, y, z, t) = \rho g(H + \eta(x, y, t) - z) \tag{6.3.1.1}$$

We conclude that the pressure gradient is indpendent of z, which means that the velocity field governing the wave motion must also be independent of z. In particular, $\partial_x u$ and $\partial_y v$ must be independent of z.

Now consider the continuity equation (2.2.0.5). Integrating the continuity equation along the z-axis over $[0, H + \eta]$ gives

$$\int_0^{H+\eta} \partial_x u + \partial_y v + \partial_z w dz = (H + \eta)\partial_x u + (H + \eta)\partial_y v + w(H + \eta) - w(0) = 0 \tag{6.3.1.2}$$

We can simplify this considerably using boundary conditions. For instance, the normal component of the fluid velocity must be zero at $z = 0$ so $w(0) = 0$. And the fluid must move with the surface at $H + \eta$ which leads to

$$w(H+\eta) - \frac{D\eta}{Dt} = \partial_t \eta + u\partial_x \eta + v\partial_y \eta \tag{6.3.1.3}$$

This allows us to write the continuity equation as

$$\partial_t + \partial_x \left[u(H+\eta)\right] + \partial_y \left[v(H+\eta)\right] = 0 \tag{6.3.1.4}$$

Dropping non-linear terms leads to the final form of the linearized shallow-water continuity equation

$$\boxed{\partial_t \eta + H\left(\partial_x u + \partial_y v\right) = 0} \tag{6.3.1.5}$$

Deriving the linearized Navier-Stokes equation is a bit trickier, so we will be skipping over some of the steps. Recall the Navier-Stokes equation for an incompressible fluid in a non-inertial reference frame (3.1.0.4)

$$\rho\frac{D\mathbf{u}}{Dt} = -\nabla p + \rho\left[\mathbf{g} - \frac{d\mathbf{U}}{dt} - 2\Omega \times \mathbf{u} - \frac{d\Omega}{dt} \times \mathbf{x} - \Omega \times (\Omega \times \mathbf{x})\right] + \mu\nabla^2\mathbf{u}$$

For waves on length scales that do are small compared to the circumferance of the Earth, the Coriolis force acceleration can be considered constant. Thus we can write

$$2\Omega \times \mathbf{u} \approx (fv, fu - 2\Omega u\cos\theta)$$

Where f is the local **Coriolis parameter** (also called the **Coriolis frequency**). Using this approximation and neglecting non-linear terms leads to

$$\partial_t u - fv = -\frac{1}{\rho}\partial_x p \tag{6.3.1.6}$$

$$\partial_t v + fu = -\frac{1}{\rho}\partial_y p \tag{6.3.1.7}$$

$$\partial_t w = -\frac{1}{\rho}\partial_z p - g \tag{6.3.1.8}$$

The final result is easily found by taking the necessary derivatives of our pressure function (6.3.1.1). Together with the continuity equation, we get the following linearized equations of motion

$$\partial_t \eta + H\left(\partial_x u + \partial_y v\right) = 0 \tag{6.3.1.9}$$

$$\partial_t u - fv = -g\partial_x \eta \tag{6.3.1.10}$$

$$\partial_t v + fu = -g\partial_y \eta \tag{6.3.1.11}$$

$$\partial_t w = 0 \tag{6.3.1.12}$$

6.3.2 Example: Kelvin waves

In the example dealing with surface gravity waves in a basin, we considered only the parts of the basin that are far from the wall, where the wave motion is exclusively in

the z-direction. We will now discuss **Kelvin waves** which are waves that arise near the walls of the basin due to Coriolis effects. Kelvin waves occur on length scales where the Coriolis parameter f can be considered constant.

The general (linearized) equations of motion for shallow-water waves subject to the Coriolis force are given by eqs. (6.3.1.9)-(6.3.1.12)

$$\frac{\partial \eta}{\partial t} + H\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right) = 0, \quad \frac{\partial u}{\partial t} - fv = -g\frac{\partial \eta}{\partial x}, \quad \frac{\partial v}{\partial t} + fu = -g\frac{\partial \eta}{\partial y}$$

Where h is the depth of the basin and η is the displacement of the wave from $z = 0$. Without loss of generality, we can consider a Kelvin wave propagating along the side of the basin that is aligned with the x-direction as in fig. 9.. In that case, $v = 0$ and the equations of motion become:

$$\frac{\partial \eta}{\partial t} + H\frac{\partial u}{\partial x} = 0, \quad \frac{\partial u}{\partial t} = -g\frac{\partial \eta}{\partial x}, \quad fu = -g\frac{\partial \eta}{\partial y}$$

The equation involving $\partial_y \eta$ shows that a displacement the y-direction is now possible due to the balance between fu and $-g\partial_y \eta$. Now choose an exponential Ansatz for the displacement and the velocity

$$\eta = \hat{\eta}(y)e^{i(kx-\omega t)} \tag{6.3.2.1}$$

$$u = \hat{u}(y)e^{i(kx-\omega t)} \tag{6.3.2.2}$$

Substituting this into the equations of motion and cancelling the exponential factors gives

$$-i\omega\hat{\eta} + ikH\hat{u} = 0, \quad -i\omega\hat{u} = -ikg\hat{\eta}, \quad f\hat{u} = -g\frac{\partial \hat{\eta}}{\partial y}$$

Isolating $\hat{u}$ in the first two equations gives the following dispersion relation:

$$\begin{aligned} \hat{u} &= \frac{\omega}{kH}\hat{\eta} = \frac{kg}{\omega}\hat{\eta} \\ &\implies \omega^2 = k^2 gH \end{aligned} \tag{6.3.2.3}$$

We see that the phase speed is:

$$c = \frac{\omega}{k} = \sqrt{gH} \tag{6.3.2.4}$$

As an aside, this agrees with the result that one finds by using the shallow-water approximation for surface gravity waves that are not affected by the Coriolis force (eq. (8.49) in Kundu). So the phase speed of Kelvin waves is the same as that of surface gravity waves.

7 Vorticity theory

First, some definitions. **Vorticity** is defined as the curl of the velocity field

$$\omega = \nabla \times \mathbf{u} \tag{7.0.0.1}$$

Lines of constant vorticity are called **vortex lines**. They are analogous to stream lines in many ways.

The **circulation** is the velocity line integral around a closed fluid loop. By Stokes' theorem

$$\Gamma = \oint_{\partial S} \mathbf{u} \cdot d\mathbf{r} = \int_S \omega \cdot \mathbf{n} dS \tag{7.0.0.2}$$

Considering the circulation around an infinitesimal area with the surface element δS (whose surface normal points in the direction of the unit vector $\mathbf{n}$) leads to

$$\mathbf{n} \cdot \omega = \frac{1}{\delta S} \oint_{\delta r} \mathbf{u} \cdot d\mathbf{r} \tag{7.0.0.3}$$

This means that at any given point, the component of ω in the direction of an arbitrary normal vector $\mathbf{n}$ is proportional to the circulation around the surrounding infinitesimal fluid loop.

See the subsection on rotational motion in the section on kinematics for a detailed discussion of the relationship between vorticity and the rotation of fluid segments.

7.1 Relation to the Coriolis effect and planetary rotation

The vorticity observed in a non-inertial frame of reference is sometimes called the **absolute vorticity** to distinguish it from the vorticity in a rotating frame. It follows the usual definition of vorticity as the curl of the velocity observed in the inertial frame

$$\omega_a = \nabla \times (\mathbf{u} + \Omega \times \mathbf{r}) = \omega + 2\Omega \tag{7.1.0.1}$$

Where Ω is the angular velocity of the rotating frame. In the context of the Earth, the term 2Ω is called the **planetary vorticity**. The component of the planetary vorticity that is normal to Earth's surface is called the **Coriolis parameter**

$$f = 2\Omega \sin\theta \tag{7.1.0.2}$$

Where θ is the latitude.

The magnitude of the component of the vorticity that is normal to Earth's surface is on the order of the magnitude of the component of the fluid velocity that is tangent to the surface, divided by the characteristic length scale for the flow

$$\omega_n \sim \left(\frac{\mathbf{u}_n}{L}\right) \tag{7.1.0.3}$$

The **local Rossby number** R_0 is defined as this quantity divided by the Coriolis frequency, and it is a measure of the strength of the local vorticity. Since the tangential component of the velocity is usually very small, this leads to the following definition

$$\frac{\omega_n}{f} = \frac{\mathbf{u}_n}{fL} = \frac{\mathbf{u}_n}{2\Omega L \sin\theta} = \frac{\varepsilon}{\sin\theta} \equiv R_0 \tag{7.1.0.4}$$

When $R_0 \ll 1$, the flow is dominated by Earth's rotation and the local vorticity is negligible. Similarly, when $R_0 \gg 1$, the flow is dominated by local vorticity effects and planetary rotation can be neglected. When $R_0 \sim 1$ both planetary effects and local vorticity must be considered.

7.2 Kelvin's circulation theorem

Kelvin's circulation theorem states:

In an inviscid, barotropic flow with conservative body forces, the circulation around a closed curve moving with the fluid remains constant with time. In other words, under those conditions

$$\frac{D\Gamma}{Dt} = 0 \tag{7.2.0.1}$$

Kelvin's circulation theorem is equivalent to the statement that there are only three ways to create or destroy vorticity in a flow:

1. Non-conservative body forces
2. A non-barotropic pressure-density relationship
3. Nonzero net viscous torques

Naturally, it follows that the vorticity of a flow will remain unaffected if the above influences are not present.

7.3 Helmholtz's vortex theorem

Helmholtz proved that the following hold for vortex motion under the restrictions that are needed for Kelvin's circulation theorem:

1. Vortex lines move with the fluid.
2. The circulation of a vortex tube (its "strength") is constant along the length of the vortex tube.
3. A vortex tube cannot end within the fluid. It must end at a boundary or form a closed loop.
4. The circulation (strength) of a vortex tube remains constant over time.

7.4 Irrotational flow / potential flow

If a flow is irrotational, meaning the vorticity is zero, then the velocity field can be written in terms of a potential (this also requires that the domain of the flow is simply-connected)

$$\mathbf{u} = \nabla\Phi \tag{7.4.0.1}$$

In this case, Bernoulli's theorem (3.5.0.4) can be used to arrive at the result

$$\partial_t\Phi + \frac{|\mathbf{u}|^2}{2} + h + \phi = \text{const.} \tag{7.4.0.2}$$

This equation is sometimes called the **integral of motion** for the fluid because it can be integrated to obtain an expression for Φ.

7.5 The vorticity equation

In an inertial frame of reference, one can write down so-called **vorticity equations** that the vorticity must obey.

For a *compressible, inviscid* fluid of constant density (i.e. one with baritropic flow), the vorticity obeys the following equation

$$\partial_t \omega + \nabla \times (\omega \times \mathbf{u}) = 0 \tag{7.5.0.1}$$

For a *incompressible, viscous* fluid with kinetic viscosity ν, it obeys the equation

$$\frac{D\omega}{Dt} = \underbrace{(\omega \cdot \nabla)\mathbf{u}}_{\text{caused by stretching/tilting of vortex lines}} + \underbrace{\nu \nabla^2 \omega}_{\text{caused by diffusion of vorticity}} \tag{7.5.0.2}$$

The viscosity of the flow allows vorticity to diffuse throughout the flow, since fluid segments can pull on nearby fluid segments through viscous friction. This viscous vorticity equation can also be written for a non-inertial frame of reference

$$\frac{D\omega}{Dt} = (\omega + 2\Omega) \cdot \nabla \mathbf{u} + \frac{1}{\rho^2} \nabla \rho \times \nabla p + \nu \nabla^2 \omega \tag{7.5.0.3}$$

7.6 The Biot-Savart law for vorticity

There is a vorticity equivalent of the Biot-Savart law from electromagnetism. It gives the velocity field when the vorticity is known and applies to incompressible flows. For a fluid volume V'

$$\mathbf{u}(\mathbf{x}, t) = \frac{1}{4\pi} \int_{V'} \frac{\omega(\mathbf{x}', t) \times (\mathbf{x} - \mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|^3} d\mathbf{x}' \tag{7.6.0.1}$$

Where the integral is over the entire volume V'.

This can be further specialized to a slender vortex element of length dl by choosing $V' = \Delta A' dl$. Then the velocity *induced by that slender vortex* is

$$\mathbf{u}(\mathbf{x}, t) = \frac{\Gamma}{4\pi} \int_{V'} \mathbf{e}_\omega(\mathbf{x}') \times \frac{\mathbf{x} - \mathbf{x}'}{|\mathbf{x} - \mathbf{x}'|^3} dl \tag{7.6.0.2}$$

Where $\mathbf{e}_\omega(\mathbf{x}')$ is the direction of the vortex element at $\mathbf{x}'$ and Γ is the circulation around the vortex (which is constant along the length of the vortex, by Helmholtz's vortex theorem).

7.7 The potential vorticity

If the following conditions hold:

- λ is a conserved quantity for each fluid element
- The frictional force is negligible

And either one of the following conditions holds:

- The fluid is barotropic, i.e. $\nabla \rho \times \nabla p = 0$

- λ is only a function of p and ρ

Then the **potential vorticity**

$$\Pi = \frac{\omega + 2\Omega}{\rho} \cdot \nabla\lambda \tag{7.7.0.1}$$

Is conserved by in each fluid element, i.e.

$$\frac{d\Pi}{dt} = 0 \tag{7.7.0.2}$$

8 Laminar Flow

Recall the Reynolds number, a dimensionless parameter that describes how strong the effects of viscous forces are compared to the inertia of the fluid

$$\mathrm{Re} \equiv \frac{\text{inertia force}}{\text{viscous force}} = \frac{\rho U L}{\mu} \tag{8.0.0.1}$$

Where U is the free-stream velocity, L is a typical length scale of the flow and μ is the viscosity.

A low Reynolds number indicates that viscous forces play a significant role in the flow and that the inviscid approximation is invalid. The following criteria applies:

- Low Re leads to *laminar*, smooth, orderly flow.
- High Re leads to *turbulent*, irregular, disordered flow.

The transition between the two regimes happens around Re $\sim$ 2000 to 3000.

The Reynolds number can also be written as $\mathrm{Re} = UL/\nu$ in terms of the kinematic viscosity $\nu = \mu/\rho$. The kinematic viscosity is related to the vorticity. For a 2D flow in the xy-plane

$$\frac{D\omega_z}{Dt} = \nu \nabla^2 \omega_z \tag{8.0.0.2}$$

Intuitively, ν measures the tendency for vorticity to diffuse throughout a fluid. This is why ν is sometimes called the *momentum diffusivity*. However, it is worth remembering that momentum transfer in a fluid flow is not strictly diffusive and advective, as can be seen from the pressure gradient term in the (incompressible) Navier-Stokes equation (2.7.0.7)

$$\frac{D\mathbf{u}}{Dt} = -\frac{1}{\rho}\nabla p + \nu \nabla^2 \mathbf{u} \tag{8.0.0.3}$$

We've dropped the term representing body forces (like gravity) since we'll only be considering the idealized flow through tubes seen from an inertial system in this section.

8.1 Example: Laminar flow between Infinite Parellel Plates

The steady laminar flow between two infinite parallel plates is one of the few situations for which an analytical solution of the Navier-Stokes equations has been found. The situation is shown in fig. 10.

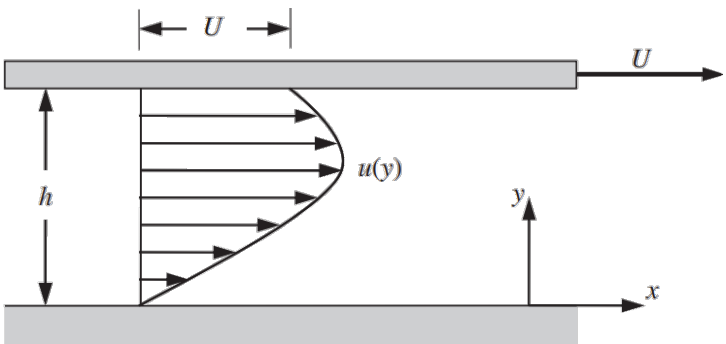


Figure 10: Steady laminar flow between two infinite parallel plates. The flow is sustained by a pressure gradient in the x-direction, as well as the motion of the upper plate with speed U in the x-direction.

Since the flow velocity in the x-direction depends only on y, we can write $\partial_x u = 0$ and apply the continuity equation (2.2.0.4) to get

$$\begin{aligned} \nabla \cdot \mathbf{u} &= \partial_x u + \partial_y v = 0 \\ \implies \partial_y v &= 0 \end{aligned}$$

Now consider the x- and y-components of the Navier-Stokes equation. Since the flow is steady, $\frac{D\mathbf{u}}{Dt} = 0$, so the x-component tells us that

$$\frac{1}{\rho}\partial_x p = \nu \partial_{yy} u = 0 \tag{8.1.0.1}$$

$$\implies \partial_x p = \text{const.} = \frac{dp}{dx} \tag{8.1.0.2}$$

While the y-component leads to

$$\frac{1}{\rho}\partial_y p = 0 \tag{8.1.0.3}$$

$$\implies p = p(x) \tag{8.1.0.4}$$

Knowing that p is only dependent on x, we can integrate the x momentum equation twice with respect to y (noting that $\rho\nu = \mu$) in order to obtain

$$0 = -\frac{y^2}{2}\frac{dp}{dx} + \mu u + Ay + B \tag{8.1.0.5}$$

For some undetermined constants A and B. To find them, we apply the boundary conditions $u = 0$ at $y = 0$ and $u = U$ at $y = h$, which leads to

$$u(y) = \frac{U}{h}y - \frac{1}{2\mu}\frac{dp}{dx}y(h - y) \tag{8.1.0.6}$$

8.2 Lubrication Theory

In general, linear or simply varying velocity profiles in a confined space lead to laminar flow. Lubrication theory is an attempt to find approximate solutions for viscous flow when the geometry is more complicated but at least one of the transverse flow dimensions is

small. Consider a 2D flow along a geometrically complex, but narrow tunnel, such as the one shown in fig. 11. The length of the passage L is assumed to be much greater than the gap height h.

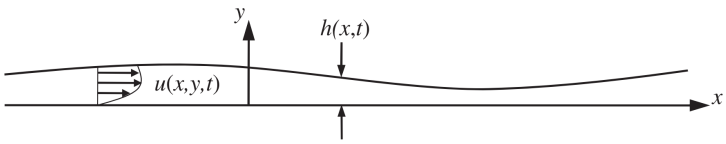


Figure 11: Viscous flow in a film bounded by a flat bottom surface and a complex upper surface. The film thickness is $h(x,t)$.

We begin by writing down the continuity equation and the momentum-equation for each component of the flow

$$\partial_x u + \partial_y v = 0 \tag{8.2.0.1}$$

$$\partial_t u + u\partial_x u + v\partial_y u = -\frac{1}{\rho}\partial_x p + \frac{\mu}{\rho}(\partial_{xx}u + \partial_{yy}u) \tag{8.2.0.2}$$

$$\partial_t v + u\partial_x v + v\partial_y v = -\frac{1}{\rho}\partial_y p + \frac{\mu}{\rho}(\partial_{xx}v + \partial_{yy}v) \tag{8.2.0.3}$$

The boundary conditions are $u = U_0(t)$ at $y = 0$ and $u = U_h(t)$ at $y = h$. We will assume that the pressure is of the form $p = p(x,t)$. We proceed to non-dimensionalize the parameters in the problem, as this will help us decide which terms can be neglected later

$$x^* = \frac{x}{L}, \quad y^* = \frac{y}{h} = \frac{y}{\varepsilon L}, \quad t^* = \frac{Ut}{L}, \quad u^* = \frac{u}{U}, \quad v^* = \frac{v}{\varepsilon U}, \quad p^* = \frac{p}{P_a} \tag{8.2.0.4}$$

Where P_a is the atmospheric pressure and $\varepsilon = h/L$ is the fineness ratio of the passage. The dimensionless form of our previous equations is

$$\partial_{x^*}u^* + \partial_{y^*}v^* = 0 \tag{8.2.0.5}$$

$$\varepsilon^2\mathrm{Re}(\partial_{t^*}u^* + u^*\partial_{x^*}u^* + v^*\partial_{y^*}u^*) = -\frac{1}{\Lambda}\partial_{x^*}p^* + \varepsilon^2\partial_{x^*x^*}u^* + \partial_{y^*y^*}u^* \tag{8.2.0.6}$$

$$\varepsilon^4\mathrm{Re}(\partial_{t^*}v^* + u^*\partial_{x^*}v^* + v^*\partial_{y^*}v^*) = -\frac{1}{\Lambda}\partial_{y^*}p^* + \varepsilon^4\partial_{x^*x^*}v^* + \varepsilon^2\partial_{y^*y^*}v^* \tag{8.2.0.7}$$

Here $\Lambda = \mu UL/P_a h^2$ is the dimensionless ratio of the viscous and pressure forces on a fluid element, called the *bearing number*. Consider the scales of the various dimensionless parameters in this problem

$$\Lambda \sim 1, \quad \mathrm{Re}\ \text{finite}, \quad \varepsilon \ll 1$$

So we can neglect terms with ε^2. Now that we know which terms can be neglected, we can go back and consider our original momentum equations, without these terms. The x and y momentum equations lead to

$$-\partial_x p + \mu\partial_{yy}u \approx 0, \quad \partial_y p \approx 0 \tag{8.2.0.8}$$

Integrating twice with respect to y leads to

$$\implies u(x,y,t) \approx \frac{1}{\mu}\partial_x p(x,t)\frac{y^2}{2} + Ay + B \tag{8.2.0.9}$$

Applying the boundary conditions $u = U_0(t)$ at $y = 0$ and $u = U_h(t)$ at $y = h$ leads to our final result for the velocity field

$$\boxed{u(x,y,t) \approx -\frac{h(x,t)}{2\mu}\partial_x p(x,t) y \left(1 - \frac{y}{h(x,t)}\right) + (U_h(t) - U_0(t))\frac{y}{h(x,t)} + U_0(t)} \tag{8.2.0.10}$$

9 Dynamic Similarity

9.1 Dimensional Analysis and Dimensionless Forms

The general principle that nature should be independent of our choice of units tells us that all laws of nature and physical relations can be written in a dimensionless form. These dimensionless forms are unitless equations governed by some dimensionless parameters specific to the problem at hand. Dimensional analysis is the tool we use to find these dimensionless parameters when we can't deduce them directly from the equations governing our system. It is the art of deriving physical relationships just by looking at units.

Before we dive into an algorithm for dimensional analysis, we need to introduce *the Buckingham theorem*, which makes the following statement:

Let $q_1, q_2, \ldots, q_n$ be variables and parameters involved in a particular physical problem with r degrees of freedom. If these variables are related by a relationship of the form $f(q_1, q_2, \ldots, q_n) = 0$, then the n variables can always be used to construct $n - r$ dimensionless parameter groups $\Pi_1, \Pi_2, \ldots, \Pi_{n-r}$, such that the relationship $f(q_1, q_2, \ldots, q_n) = 0$ can be rewritten as a dimensionless relationship

$$\phi(\Pi_1, \Pi_2, \ldots \Pi_{n-r}) = 0, \quad \text{or} \quad \Pi_1 = \varphi(\Pi_2, \Pi_3, \ldots, \Pi_{n-r}) \tag{9.1.0.1}$$

We are now ready to proceed with *Rayleigh's algorithm* for constructing the dimensionless relationship.

1. Select the variables/parameters of interest $q_1, q_2, \ldots, q_n$. Assume they are related via some relationship $f(q_1, q_2, \ldots, q_n) = 0$.
2. Construct the dimensional matrix. Each row corresponds to a dimension in your unit system, e.g. $[M]$ or $[L]$. Each column corresponds to one of the q_n.
3. Determine the rank r of the dimensional matrix. This is the number of degrees of freedom in the system.
4. Determine the number of dimensionless groups, $n - r$.
5. Construct the dimensionless groups. This can be done by exponent algebra. First select r parameters $q'_1, q'_2, \ldots, q'_r$ that will be used to construct all of the dimensionless groups. Then the i-th dimensionless group (corresponding to the i-th unused parameter that was not selected as one of the r parameters) can be written as a product of q_i and q'-parameters raised to unknown powers $a, b, c, ...$

   $$\Pi_i = q_i q_1'^a q_2'^b q_3'^c ...$$

   Use dimensional analysis to ensure that the Π_i-group is dimensionless by choosing $a, b, c, ...$ appropriately. Repeat this $n-r$ times with all of the parameters that were not chosen as q'-parameters until you have constructed $n-r$ dimensionless groups. TIP: In fluid mechanics, it's often a good idea to choose the q' to be a characteristic velocity, characteristic length and a fluid property involving mass, if possible.
6. State the dimensionless relationship. To do so, write down a relationship of the form presented in eq. (9.1.0.1).

7. Use physical intuition/knowledge to simplify the dimensionless relationship if possible.

9.2 Dynamical Similarity

The principle of *dynamical similarity* can be formulated as:

If two systems are geometrically similar (e.g. a model of a wing with the exact same shape as the real wing) and if each independent dimensionless parameter for the model system is equal to the corresponding independent dimensionless parameter for the true system, then the dependent dimensionless parameters for the model and true system will be equal as well.

In other words, two systems are dynamically similar if they are geometrically identical (except for scale) and have the same dimensionless parameters, in which case the fluid dynamics of the two systems will be identical.